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Slings

Autologous, Biologic, Synthetic, and Midurethral

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Keywords

complications; midurethral sling; outcomes; pubovaginal sling; stress urinary incontinence; surgery

Questions

1. The Integral Theory proposed by Petros and Ulmsten states that:
    a. urethral hypermobility is the primary cause of stress urinary incontinence (SUI).
    b. adequate function of the pubourethral ligaments, the suburethral vaginal hammock, and the pubococcygeus muscle helps to preserve continence.
    c. anchoring a sling to the rectus muscle allows it to respond to changes in intra-abdominal pressure.
    d. dynamic kinking of the urethra during stress preserves incontinence.
    e. synthetic mesh integrates into host tissue.
2. Which of the following statements about the preoperative assessment of a sling patient is TRUE?

a. It is generally not necessary to perform a focused neurologic examination.
b. Urgency is not associated with worse outcomes after sling surgeries.
c. The American Urological Association (AUA) Guidelines state that a postvoid residual (PVR) volume should be checked on all patients.
d. Cystoscopy should be performed in all patients to rule out bladder pathology.
e. The abdominal leak-point pressure is traditionally defined by a pressure of less than 80 cm H_2O.

3. As an option for the index patient with SUI, the AUA SUI guidelines strongly recommends that physicians offer:
   a. synthetic MUS.
   b. autologous fascia pubovaginal sling (PVS).
   c. Burch colposuspension.
   d. bulking agents.
   e. all of the above.
4. Which of the following statements regarding PVS materials is TRUE?
   a. Harvesting a thin strip of fascia lata is associated with significant morbidity.
   b. The risk of perforation and exposure associated with synthetic slings is minimal.
   c. The benefit of harvesting fascia lata is that patient position can be maintained for the entire case.
   d. Synthetic sling materials exhibit the least amount of degradation.
   e. Although concerning, disease transmission has never been documented with allograft materials.
5. An autologous PVS is indicated in all of the following conditions EXCEPT:
   a. urethral incompetence in a T12 spinal cord injury.
   b. low urethral resistance with decreased bladder compliance.
   c. urethral incompetence and large urethral diverticulum.

d. proximal urethral loss secondary to long-standing indwelling Foley catheter.
e. refractory SUI after failed mid-urethral sling (MUS) and bulking agents.

6. Which of the statements about the normal female urethra and pelvic floor is TRUE?
    a. The female urethra is composed of four separate tissue layers, and the middle seromuscular layer is most important in enhancing the urethral sphincter mechanism during voiding.
    b. The Valsalva pressure of the bladder exceeds the resting closing pressure of the internal sphincter.
    c. The fast-twitch fibers of the external sphincter are responsible for sudden protection against incontinence, and slow-twitch fibers provide passive control through the involuntary guarding reflex.
    d. The levator ani, urethropelvic ligament, and round ligament provide needed support to the bladder neck and undersurface of the bladder.
    e. The PVS is placed at the bladder neck to provide adequate urethral coaptation at rest and to decrease urethral responsiveness to abdominal pressure.
7. Which of the following statements regarding materials for bladder neck PVSs is FALSE?
    a. The ideal material has minimal tissue reaction and complete biocompatibility.
    b. Stiffness and maximal load failure are the same between freeze-dried fascia lata and solvent-dehydrated and dermal grafts.
    c. The estimated risk of human immunodeficiency virus (HIV) transmission by an allograft sling is approximately 1 in 1,660,000.
    d. Porcine small intestinal submucosa has less tensile strength than cadaveric fascia lata.
    e. Synthetic materials are associated with high perforation rates during use for bladder neck PVS.
8. Before final tensioning of the rectus fascial autologous PVS:

a. the weighted speculum is removed.
b. the abdominal skin incision should be closed.
c. vaginal packing should be placed.
d. the patient should be taken out of lithotomy position.
e. a drain should be placed in the retropubic space.

9. In outcomes associated with PVS procedures, which of the following is/are TRUE?
    a. Reported cure rates after an autologous PVS procedure are 24% to 97%.
    b. Preoperative Valsalva leak point pressure is a reliable predictor of outcomes after sling surgery.
    c. In the Stress Incontinence Surgical Treatment Efficacy Trial (SISTEr) trial, cure rates and voiding symptoms were greater for the PVS than for the Burch colposuspension.
    d. a and c.
    e. a, b, and c.
10. Which of the following statements about perforation and PVS material is TRUE?
    a. Synthetic slings perforate into the urinary tract 15 times more often than autologous, allograft, or xenograft slings.
    b. Urethral perforations are rarely associated with urinary retention and mixed urinary incontinence.
    c. Synthetic slings are less likely to be associated with vaginal exposure than autologous, allograft, and xenograft slings.
    d. Perforation from synthetic slings requires removal of the entire sling from a vaginal and retropubic approach.
    e. None of the above are true.
11. Which of the following statements is NOT associated with voiding dysfunction after a PVS procedure?
    a. Obstruction, detrusor overactivity, or impaired detrusor contractility are all manifestations of voiding dysfunction for iatrogenic PVS obstruction.
    b. Persistent urgency is more common than urinary retention in bladder outlet obstruction after a PVS procedure.
    c. Fifty percent of affected patients have symptoms of overactive bladder, which can be avoided if sling lysis is performed

within 2 weeks of PVS placement.
    d. Urodynamic study is valuable in assessment and planning management.
    e. There is up to a 20% recurrent SUI rate after urethrolysis.

12. Regarding the pathophysiology of incontinence:
    a. hypermobility is the main underlying cause of SUI.
    b. intrinsic sphincter deficiency (ISD) is rarely the primary cause of SUI.
    c. ISD is the primary underlying cause of SUI for women, with hypermobility being a secondary finding.
    d. the levator floor provides active compression to the proximal urethra.
    e. the extrinsic urethral skeletal sphincter is the primary mechanism for urinary continence.
13. Obese patients who undergo MUS surgery:
    a. clearly have a higher rate of sling-related complications.
    b. should have been offered weight loss as an initial management option.
    c. have a significantly higher risk of trocar injury at the time of sling placement.
    d. have been consistently shown to have worse outcomes.
    e. are at a greater risk for voiding dysfunction.
14. All of the following are associated with the MUS procedure EXCEPT:
    a. trocars are inserted and used to transpose the implanted material into position.
    b. the synthetic material used is a mesh with large pores.
    c. loose tension is placed on the sling material.
    d. the sling is sutured to the underlying tissues for fixation purposes.
    e. cystoscopy is a crucial component of the procedure.
15. Common presenting symptoms of voiding dysfunction after PVS surgery are:
    a. urgency and frequency.
    b. painful voiding and suprapubic pain.
    c. incomplete emptying and straining.

d. recurrent urinary tract infections.
e. all of the above.

16. The transobturator MUS passes through all of the following structures EXCEPT the:
    a. obturator internus muscle.
    b. obturator membrane.
    c. obturator externus muscle.
    d. rectus fascia.
    e. ischiorectal fascia.
17. In review of the efficacy outcomes obtained with mid-urethral procedures, which of the following is TRUE?
    a. MUSs are less effective than open colposuspension procedures.
    b. MUSs produce inferior results compared with laparoscopic colposuspensions.
    c. Postoperative voiding dysfunction is more common with mid-urethral procedures than with other types of suspension procedures.
    d. Mixed urinary incontinence results are superior to those for pure SUI.
    e. Five-year results demonstrate durability similar to one-year results.
18. Which of the following is theorized to be TRUE regarding patients at risk for voiding dysfunction after PVS surgery?
    a. Failure to relax the external striated sphincter is not associated with postoperative voiding dysfunction.
    b. Patients who habitually void with abdominal straining will not have an increased risk of voiding dysfunction after PVS surgery.
    c. Patients with pure SUI are more likely to have voiding dysfunction after PVS surgery.
    d. Patients with subclinical impaired detrusor contractility are at increased risk for voiding dysfunction after PVS surgery.
    e. Young patients are more at risk for voiding dysfunction after PVS surgery.
19. In placing the single-incision mini sling, the passer punctures the:

a. obturator internus muscle.
b. obturator membrane.
c. obturator externus muscle.
d. rectus fascia.
e. ischiorectal fascia.

20. Compared to younger patients, MUSs in elderly patients are associated with:
    a. lower success rates.
    b. lower satisfaction rates.
    c. higher rates of postoperative urgency.
    d. higher rates of mixed incontinence resolution.
    e. higher rates of postoperative urinary retention.
21. In a patient with voiding dysfunction after sling surgery:
    a. it is generally appropriate to wait as long as 3 months after MUS surgery before considering surgical intervention.
    b. it is generally appropriate to wait as long as 3 months after autologous PVS surgery before considering surgical intervention.
    c. loosening a synthetic sling through traction with a cystoscope in the operating room is associated with little risk.
    d. formal urethrolysis has been shown to be superior to sling incision.
    e. intermittent catheterization is not advisable.
22. When MUSs are performed at the time of prolapse surgery:
    a. risks of perforation, exposure, and infection are higher than in cases in which only a sling is performed.
    b. concomitant hysterectomy has an adverse effect on incontinence outcome.
    c. rates of urethrolysis for postoperative retention are higher.
    d. occult incontinence may not be adequately addressed.
    e. rates of retention are slightly higher than in those undergoing a sling procedure only.
23. When MUSs are used as salvage procedures:
    a. complication rates are higher than when MUSs are done primarily.

b. the technique needs to be altered compared to a primary procedure.
c. failure rates are unaffected by urethral hypermobility.
d. bladder perforation is less than in primary cases.
e. overall efficacy is similar to that of primary implantation.

24. Complications associated with MUSs include:
    a. bladder perforation in up to 5%.
    b. voiding dysfunction in 4% to 20%.
    c. de novo urgency in as many as 12%.
    d. delayed wound healing in approximately 1%.
    e. all of the above.
25. According to International Continence Society (ICS) and International Urogynecological Association (IUGA) terminology pertaining to synthetic (prosthetic) mesh sling complications, which of the following is TRUE?
    a. The term *perforation* should be used when mesh is present within the urinary tract or bowel.
    b. The term *exposure* should be used when mesh is present in the urinary tract or bowel.
    c. The term *erosion* should be used when mesh is found in the urinary tract a year or more after surgery.
    d. a and b.
    e. a and c.
26. Material-related exposures and perforations associated with MUSs are:
    a. decreased by the macroporous nature of the sling material.
    b. unaffected by tension placed on the slings.
    c. associated with vaginal exposures approximately 20% of the time.
    d. associated with bladder perforation rates of 20%.
    e. irrelevant to outcomes or satisfaction.
27. Which of the following statements about the anatomy of MUSs is TRUE?
    a. The obturator nerve and vessels are less than 2 cm away from the transobturator sling at the level of the obturator foramen.

b. For retropubic slings, the dorsal nerve of the clitoris is typically less than 2 cm away from the sling.
c. The anatomic position of a single-incision sling is significantly affected by position of the legs.
d. A branch of the obturator artery that courses along the pubic bone is more likely to be injured with an in-to-out transobturator sling technique.
e. The periurethral fascia covering the posterior urethra is very thin.

28. Which of the following statements is FALSE regarding the treatment of patients with recurrent SUI with a MUS surgery?
    a. Retropubic slings have been shown to have better outcomes than transobturator slings in patients with recurrent SUI in a few small series.
    b. Repeat MUS surgery is significantly less effective at curing incontinence than primary MUS surgery.
    c. A meta-analysis of MUS surgery for recurrent SUI found that retropubic MUS surgery was significantly better than transobturator MUS surgery.
    d. Recurrent SUI after MUS surgery may be due to ISD.
    e. None of the above statements are false.
29. In regard to perforations associated with MUSs:
    a. bladder perforations cannot be managed endoscopically in well-selected cases.
    b. vaginal exposures cannot be managed conservatively.
    c. exposures and perforations are not related to errant sling placement.
    d. symptoms are not usually associated with exposure.
    e. complete excision of exposed material should be performed.
30. In regard to the mechanics of MUSs, which of the following is TRUE?
    a. MUSs work primarily by compressing the urethra.
    b. There is no evidence to support dynamic kinking of the urethra as a mechanism for continence for MUSs.
    c. Placing a sling tight at the mid-urethra will help eliminate postoperative hypermobility.

d. It appears that a MUS works by impeding the movement of the posterior urethral wall.
e. Postoperative urethral hypermobility is associated with failure of the procedure.

31. Which of the following statements is TRUE?
    a. Retropubic MUSs cure SUI better than transobturator MUSs.
    b. The risk of urinary tract trocar injury is higher with retropubic MUSs than transobturator MUSs.
    c. Postoperative voiding dysfunction is higher with transobturator MUSs than retropubic MUSs.
    d. It is not necessary to perform cystoscopy after a transobturator MUS surgery.
    e. It is not necessary to perform cystoscopy after a single-incision MUS surgery.
32. Voiding dysfunction associated with MUSs is:
    a. not associated with changes in urodynamic parameters.
    b. predictable based on unique preoperative voiding parameters such as flow rate.
    c. managed by immediate sling release.
    d. managed initially conservatively, but sling release should be contemplated when persistent voiding trials are not successful.
    e. resolved by complete excision of the sling.
33. In regard to operative management for voiding dysfunction after MUS surgery:
    a. a single incision of the sling results in incontinence in the majority of patients.
    b. it is important to remove the entire sling.
    c. surgery should not be considered until at least 3 months after sling placement as after autologous PVSs,.
    d. voiding dysfunction is usually transient.
    e. loosening the sling in the operating room with a cystoscope is a very safe option.
34. Complications associated with MUSs include:
    a. superficial vaginal material exposure.
    b. vascular perforation.
    c. intestinal perforation.

d. significant hemorrhage requiring transfusion.
e. all of the above.

35. Which of the following statements is TRUE regarding sexual dysfunction after MUS surgery?
    a. Postoperative dyspareunia is not associated with MUS surgery.
    b. Sling removal has been shown to improve dyspareunia.
    c. It has been clearly shown that MUS surgery will improve the sexual function of a woman with incontinence.
    d. A decrease in coital incontinence may improve sexual function.
    e. b and d.
36. Which of the following statements is TRUE regarding bleeding and hematomas after MUS surgery?
    a. The rate of undiagnosed hematomas is likely less than 5%.
    b. The majority of postoperative hematomas resolve without intervention.
    c. In the literature, the rate of severe bleeding is consistently less than 1%.
    d. The rate of hematomas and severe bleeding is lower after retropubic MUS surgery than transobturator MUS surgery.
    e. All of the above are true.
37. Regarding the transobturator technique, the:
    a. surgical placement of the tape requires insertion through the adductor longus tendon.
    b. tape never traverses the gracilis or adductor magnus brevis muscles.
    c. anterior branch of the obturator artery is located at the medial aspect of the obturator foramen.
    d. tape remains above the perineal membrane and outside the true pelvis and does not penetrate the levator ani group.
    e. dorsal nerve of the clitoris is in close juxtaposition to the tape.
38. The transobturator technique involves:
    a. either outside-in or inside-out approaches.
    b. no absolute requirement for cystoscopy.
    c. no risk of lower urinary tract injury.
    d. no risk of leg pain or dyspareunia.

e. similar meshes in all available kits.

39. Reported outcomes with the transobturator MUS:
    a. appear to be relatively similar regardless of whether ISD is present preoperatively.
    b. include bladder, but not urethral, injury being reported.
    c. indicate that vaginal exposure is similar regardless of the type of tape used.
    d. show that voiding dysfunction is significantly less with this technique as compared with the retropubic approach.
    e. are not affected by the presence of urethral hypermobility.
40. Which of the following statements is TRUE regarding pain after MUS surgery?
    a. Groin pain is more commonly associated with the transobturator MUS approach.
    b. When groin pain does occur, it persists longer after retropubic MUS surgery.
    c. Most groin pain resolves after 2 days.
    d. a and c.
    e. a and b.
41. Which of the following statements is TRUE regarding infection after MUS surgery?
    a. Severe infection is a common complication after MUS surgery.
    b. Randomized controlled trials demonstrate a vaginal wound infection rate of approximately 10%.
    c. There are no reports in the literature about delayed presentation of infection after MUS surgery.
    d. Obesity and diabetes are associated with fasciitis after pelvic surgery.
    e. All of the above are true.
42. Which of the following is FALSE regarding the surgical management of mesh perforation and exposure after MUS surgery?
    a. Observational treatment is not recommended for mesh perforation of the bladder.
    b. Endoscopic excision or ablation is an acceptable first step for select patients with small areas of bladder perforation.

c. A midline vaginal incision is acceptable for patients undergoing removal of mesh that has perforated into the urethra.
d. Reconstruction should involve nonoverlapping suture lines and may benefit from interposition of tissue such as a labial fat pad, greater omentum, or autologous fascial sling.
e. In cases of mesh perforation of the urinary tract, the adjacent MUS portions outside the urinary tract should also be removed.

43. Which of the following is FALSE regarding recurrent or persistent SUI after MUS?
    a. Persistent SUI may be due to poor surgical technique, wrong choice of procedure, or suboptimal patient selection.
    b. Cystoscopy and urodynamics are indicated in the workup of recurrent SUI.
    c. Sling tightening is not supported by evidence-based literature.
    d. Periurethral bulking produces inferior outcomes to repeat MUS for recurrent SUI.
    e. The data support the use of autologous PVS over repeat MUS in this population.
44. Which of the following statements is TRUE regarding regulatory and legal issues related to sling mesh complications?
    a. The first MUS had to go through the premarket approval (PMA) process, and then subsequent slings were approved through the 510(k) process.
    b. MUSs can no longer use the 510(k) approval process.
    c. The FDA considers mesh complications to be “rare.”
    d. In the legal profession, the § symbol does NOT stand for “section.”
    e. Single-incision mini-sling manufacturers are required to perform 522 postmarket surveillance studies.
45. In 2016, the following medical devices have been reclassified as class III:
    a. retropubic midurethral slings.
    b. single-incision mini-slings.
    c. surgical mesh for transvaginal prolapse.

d. surgical mesh for sacral colpopexy.
e. b and c.

Answers

1. **b. Adequate function of the pubourethral ligaments, the suburethral vaginal hammock, and the pubococcygeus muscle helps to preserve continence.** They postulated that injury to any of these three components from surgery, parturition, aging, or hormonal deprivation could lead to impaired mid-urethral function and subsequent urinary incontinence.
2. **c. The American Urological Association (AUA) Guidelines state that a postvoid residual (PVR) volume should be checked on all patients. Based on AUA guidelines, a urinalysis and measurement of PVR volume should be performed on all patients, but more extensive imaging is not part of the routine evaluation of urinary incontinence.** However, in some patients abnormal findings in the history, physical examination, or urinalysis may warrant further imaging.
3. **e. All of of above. All of the following are options for the index patient and are supported by Grade A evidence.** The choice of intervention should be individualized based on the patient's symptoms, the degree of bother the symptoms cause the patient, patient goals and expectations, and the risks and benefits for a given patient.
4. **d. Synthetic sling materials exhibit the least amount of degradation.** In 2008, Woodruff et al. [a] examined explanted sling materials and determined that synthetic materials demonstrated the least amount of degradation. They also demonstrated the greatest amount of fibroblast and tissue ingrowth into the specimen.
5. **b. Low urethral resistance with decreased bladder compliance.** Decreased bladder compliance is of concern for upper tract deterioration. The addition of a PVS, by increasing bladder outlet resistance, would cause significant damage to the upper tracts. The compliance should be addressed before or concurrently to anti-

incontinence measures. **A PVS procedure is indicated for ISD associated with urethral hypermobility, SUI presenting as concomitant cystoceles, SUI associated with urethral diverticulum, urethral damage from previous MUS, SUI associated with urethral defects (e.g., urethrovaginal fistula) in which urethral reconstruction is required, and in women with SUI and associated neurogenic conditions.**

6. **c. The fast-twitch fibers of the external sphincter are responsible for sudden protection against incontinence, and slow-twitch fibers provide passive control through the involuntary guarding reflex.** The female urethra is composed of four layers, with the middle muscular layer maintaining the resting urethral closure mechanism and the outer seromuscular layer augmenting this closing pressure. The levator ani, urethropelvic ligament, and pubocervical fascia provide support to the bladder neck and underside of the bladder. The round ligament provides support to the uterus. **A PVS is placed at the bladder neck to provide adequate urethral coaptation for increasing urethral responsiveness to abdominal pressure.**
7. **b. Stiffness and maximal load failure are the same between freeze-dried fascia lata and solvent-dehydrated and dermal grafts.** Maximum load to failure, maximum load/graft width, and stiffness are significantly lower for the freeze-dried fascia lata group compared with the autologous, solvent-dehydrated, and dermal graft groups. The ideal graft material causes no tissue reaction, is completely biocompatible, leads to significant host fibroblast infiltration and neovascularization, and causes negligible perforation or exposure. The estimated risk of HIV transmission from an allograft is 1 in 1,667,600. The theoretical risk of developing Creutzfeldt-Jakob disease from a non-neural allograft is 1 in 3.5 million. Porcine small intestinal submucosa has less tensile strength than cadaveric fascia lata. Synthetic material is no longer used for bladder neck PVS because of the exceedingly high perforation rates.
8. **a. The weighted speculum should be removed.** Tensioning before the speculum is removed may result in failure of the procedure due

to too much or too little tension. The abdominal incision is also closed after the sling is tensioned.

9. **d. a and c.** Cure rates reported in peer-reviewed literature for autologous PVS procedures are 24% to 97%. There are no risk factors that predict outcomes after PVS surgery for primary or recurrent SUI. The SISTEr trial was a multicenter, randomized clinical trial (Albo et al., 2007[a]) that found higher cure rates for the PVS procedure than the Burch colposuspension, but also more associated voiding symptoms (urinary tract infection, difficulty voiding, and postoperative urge incontinence, $P < .001$).
10. **a. Synthetic slings perforate into the urinary tract 15 times more often than autologous, allograft, or xenograft slings.** This includes sutures, bone anchors, and screws. **Synthetic slings perforate 15 times more often into the urethra and are exposed 14 times more often into the vagina than autologous, allograft, and xenograft slings.** Urethral perforation usually presents at a mean of 9 months as urinary retention, urgency, and mixed urinary incontinence.
11. **c. Fifty percent of affected patients have symptoms of overactive bladder, which can be avoided if sling lysis is performed within 2 weeks of PVS placement. Transient urinary retention is common, and most patients return to spontaneous voiding within 10 days postoperatively.** Obstructive symptoms may improve or resolve with time, which is the reason most physicians prefer waiting 3 months before considering surgical intervention. The incidence of voiding dysfunction after continence surgery varies from 2.5% to 35% and includes obstruction, detrusor overactivity, or impaired detrusor contractility. **Persistent postoperative urgency incontinence and urgency present more commonly (8% to 25%) than frank retention.** Although urodynamics do not preoperatively predict outcomes after anti-incontinence surgery or urethrolysis, it is useful in diagnosing and treating patients with obstruction after a PVS procedure. There is a 0% to 18% recurrent SUI rate after urethrolysis.

12. **c. ISD is the primary underlying cause of SUI for women, with hypermobility being a secondary finding.** Although urethral hypermobility is present in many women, most do not manifest incontinence, and therefore, ISD is considered to be the most important factor in women who experience urinary loss. The extrinsic urethral sphincter is not considered to be the primary mechanism for urinary continence in women. The ongoing debate regarding hypermobility and ISD is further compounded by the advent of midurethral slings, which clearly address hypermobility during stress events. Given the efficacy of midurethral slings, there has been some confusion regarding the role of hypermobility in promoting continence. However, most believe that the intrinsic urethral mechanism is of primary importance for urinary control.
13. **b. Should have been offered weight loss as an initial management option. It has been consistently shown in the literature that obese patients with incontinence benefit from weight loss.** The literature regarding improvement or cure of incontinence in patients with obesity compared with nonobese patients is mixed. Multiple authors have found a higher rate of bladder trocar injury in nonobese patients during MUS surgery.
14. **d. The sling is sutured to the underlying tissues for fixation purposes.** The tension-free vaginal tape (TVT) procedure incorporates several specific technical components. Insertion trocars are used in either a suprapubic or a vaginal approach to assist in implantation of the material in the retropubic area. It is now well known that type 1 synthetic meshes are best because of their wide porosity. In addition, this mesh should be monofilamentous. **Most authorities recommend placement of loose tension only on the TVT, although some authorities now are placing greater tension on the TVT, with success being established in patients with lesser degrees of hypermobility.** No suture fixation to the underlying periurethral fascia is necessary to anchor the sling. Cystoscopy is a vital component of this procedure to exclude urinary tract injury.
15. **e. All of the above.** The presentation of patients with voiding dysfunction is variable, and the symptoms range from complete

urinary retention and urgency incontinence to the less obvious irritative symptoms. Obstruction may also present with recurrent urinary tract infections, prolonged suprapubic pain, and painful voiding, even if emptying is completed.

16. **d. Rectus fascia.** During placement of the transobturator MUS, the trocar must traverse the obturator internus muscle, obturator membrane, and obturator externus muscle as it goes through the obturator foramen. Medially, the transobturator MUS enters the anterior compartment of the ischiorectal fascia, in the area of the levator ani membrane and obturator internus muscle. While the MUS passes through the rectus fascia, the transobturator sling does not.
17. **e. Five-year results demonstrate durability similar to 1-year results.** Five-year (and now 7-year) longitudinal results have shown that midurethral slings have procedural durability in terms of efficacy. This efficacy is not substantially less than results obtained at 1 year. Furthermore, several studies with greater than 10-year outcomes have recently been published and reveal similar durability in the longer term. Randomized trials have demonstrated similar efficacy in patients undergoing either open colposuspensions or laparoscopic colposuspensions. MUSs provide superior results compared with laparoscopic procedures. Although voiding dysfunction may be observed after any type of sling procedure, results suggest that midurethral slings are associated with less voiding dysfunction than either colposuspensions or bladder neck slings. Results with mixed incontinence are acceptable compared with other types of interventions for urinary incontinence but are less than those obtained in pure SUI.
18. **d. Patients with subclinical impaired detrusor contractility are at increased risk for voiding dysfunction after PVS surgery. It has been shown that preoperative voiding dysfunction affects a patient's ability to empty after anti-incontinence surgery.** Subclinical preoperative impaired detrusor contractility may manifest symptomatically with voiding dysfunction after PVS surgery. Dysfunctional voiding or failure of relaxation of the external urethral sphincter may also affect emptying after surgery.

Also, a patient who habitually voids by abdominal straining may have difficulty emptying after incontinence surgery. Because of the variability of presenting symptoms following a PVS, it is important to ascertain the predominant symptom with a thorough history.

19. **a. Obturator internus muscle.** None of the other structures should be penetrated during the procedure.
20. **c. Higher rates of postoperative urgency. They are associated with rates of postoperative urgency that are higher than those in young patients.** Elderly patients experience higher rates of postoperative urgency associated with any sling material, and this is true for the MUS as well. However, elderly patients have results similar to their younger peers, and therefore satisfaction rates are also similar to those of their younger peers. Mixed urinary incontinence resolution rates are similar to those of the younger population, and actual postoperative retention occurs to a similar degree as in younger patients, but postoperative voiding function may be slightly higher in the older population.
21. **b. It is generally appropriate to wait as long as 3 months after autologous PVS surgery before considering surgical intervention.** Obstruction following an autologous PVS usually improves or resolves with time, therefore, most physicians historically have preferred waiting 3 months before considering surgical intervention after PVS (it may not be suitable to wait this long after midurethral slings). It is appropriate and effective to initially treat persistent voiding dysfunction conservatively. This includes temporary catheter drainage, clean intermittent catheterization, timed voiding, double voiding, biofeedback, pelvic floor muscle training, and anticholinergic therapy.
22. **d. Occult incontinence may not be adequately addressed.** Midurethral slings performed at the time of prolapse surgery have now been shown to be safe and efficacious; however, appropriate MUS tensioning in women with occult SUI (only seen with POP reduction) may be challenging. Risks of perforation, exposure, and infection are no greater than when the midurethral sling is performed as a primary isolated procedure. Concomitant hysterectomy has been shown not to have an adverse effect on

continence status associated with these procedures. In addition, rates of postoperative urethrolysis are no greater when the midurethral sling technology is combined with a prolapse correction. Rates of retention are also not appreciably higher in this population, compared with those women undergoing isolated slings only.

23. **e. Overall efficacy is similar to that of primary implantation.** As salvage procedures, midurethral slings have overall efficacy similar to their use in primary implantation procedures. Complications should be no higher than when done as primary procedures. The technique remains the same, and no alteration is required. Success does appear to be reliant on hypermobility, and patients with less hypermobility would appear to have less overall functional success than those patients with greater hypermobility. Rates of bladder perforation may be somewhat higher in this population than in primary cases.
24. **e. All of the above.** Complications with midurethral slings are an important part of informed consent. Bladder perforation rates range as high as 5% and in some studies are somewhat higher. Voiding function rates vary from 4% to 20%, and this variance is largely related to definitional reasons based on literature evidence. **De novo urgency occurs with postoperative voiding dysfunction in as many as 12% of patients, and wound healing can be affected in approximately 1% of patients; results represent dramatic improvement compared with historic dense weave meshes.**
25. **d. a and b.** In 2010, the IUGA and the ICS released a report clarifying and standardizing the terminology related to complications from insertion of synthetic and biological materials during female pelvic surgery (Haylen et al., 2011[a]). According to that report, synthetic mesh is termed a *prosthesis* and a biologic implant is termed a *graft.* Mesh located in the lower urinary tract is termed a *perforation* and extrusion of mesh through the skin or vagina is termed *exposure.*
26. **a. Decreased by the macroporous nature of the sling material. Exposure and perforation associated with midurethral slings is**

clearly decreased by the use of macroporous monofilament sling material (type 1). Tension may have a role in increasing mesh-related complications even in macroporous slings. **Vaginal exposure rates and bladder perforation rates are very low and do not exceed 5% to 10% with newer sling materials.** When material complications do occur, however, they have an adverse impact on overall patient satisfaction.

27. **b. For retropubic slings, the dorsal nerve of the clitoris is typically less than 2 cm away from the sling.** The left and right dorsal nerves of the clitoris (DNC) run along the inferior surface of the ischiopubic rami and cross under the pubic bone approximately 1.4 cm from the midline. Therefore, when a placing a trocar, it is important to stay at least 2 cm from the midline to avoid injuring the DNC.
28. **c. A meta-analysis of MUS surgery for recurrent SUI found that retropubic MUS surgery was significantly better than transobturator MUS surgery.** In 2013, Agur et al.[a] performed a meta-analysis of the 10 randomized, controlled trials that of MUSs that addressed recurrent SUI. The review included 350 women with a mean follow-up of 18.1 months. The authors found no significant difference in subjective cure rates in patients after retropubic versus transobturator MUS surgery.
29. **e. Complete excision of exposed material should be performed.** Management of exposures and perforations is complex and must be individualized. **Primarily, all exposed material, whether it be vaginal or within the urinary tract, must be removed or in some manner covered.** There have been successful reports of bladder management endoscopically, although this is contingent on absolute excision of all exposed material. Some authors have reported successful management of vaginal exposures with conservative use of topical estrogens and delayed primary closure as well as simple secondary intention healing. Exposures and perforations are clearly linked to technique, and errant sling placement has a high significance in creating perforations.

30. **d. It appears that a MUS works by impeding the movement of the posterior urethral wall.** Indeed, it appears that a midurethral sling works by impeding the movement of the posterior urethral wall above the sling, directing its motion in an anteroinferior or anterior direction. In addition, inward movement of the posterior urethral wall after placement of a midurethral sling results in urethral lumen narrowing (compression). This securing of the posterior wall of the urethra (with or without compression during stress maneuvers) is one theory of how midurethral slings achieve continence.
31. **b. The risk of urinary tract trocar injury is higher with retropubic MUSs than transobturator** MUS**s.** In the majority of published series comparing retropubic and transobturator midurethral slings, the rate of urinary tract trocar injury at the time of sling placement is higher with retropubic slings. However, there are numerous case reports of transobturator sling mesh perforating into the urinary tract. Therefore, cystoscopy should be performed after transobturator sling trocar passage.
32. **d. Managed initially conservatively, but sling release should be contemplated when persistent voiding trials are not successful.** Voiding dysfunction associated with midurethral slings is substantially less than with bladder neck slings but still occurs. Timing of intervention is dependent on surgeon experience but is trending toward earlier intervention. Most experts recommend a period of conservative management of a few days to 1 month. Persistent obstruction will require intervention. Urodynamic parameters are often affected in cases of persistent obstruction. Unfortunately, no preoperative factors are predictive of postoperative voiding dysfunction. Immediate release is not recommended because a short period of observation usually results in resolution of the voiding dysfunction. **When sling release occurs, midline incision of the sling is all that is required; the entire sling does not need to be excised.**
33. **d. Voiding dysfunction is usually transient.** Urinary obstruction after MUS surgery is usually transient and can be managed with short-term intermittent catheterization, although occasionally

symptoms mandate sling release. Long-term retention after retropubic midurethral sling surgery is a rare complication. In these cases, removal or incision of the sling usually improves the patient's symptoms.

34. **e. All of the above.** Complications of technique include injury to surrounding structures and significant hemorrhage due to laceration of perivesical vessels. Intestinal and vascular complications can cause substantial morbidity and mortality.
35. **e. b and d.** The rate of de novo dyspareunia after MUS surgery is between 3% and 14.5%. Some authors attribute improved sexual function after MUS surgery to a significant decrease in coital incontinence. There is contradictory evidence in the literature that MUS surgery improves and worsens sexual function. There is evidence in the literature that sling removal can improve dyspareunia.
36. **b. The majority of postoperative hematomas resolve without intervention.** In 2005, Tseng et al.[a] performed ultrasound on 62 women after MUS surgery and found that 8 (12.9%) patients had significant retropubic hematomas larger than 5 cm on the day after surgery. Repeat ultrasonographic examinations 1 month after surgery revealed all the hematomas except one had resolved. In randomized controlled trials, the majority of bleeding complications occur in patients after retropubic MUS surgery.
37. **d. Tape remains above the perineal membrane and outside the true pelvis and does not penetrate the levator ani group.** The transobturator technique is unique because (when done correctly) it avoids entry into the true pelvis and the levator group. Errant sling placement through the adductor longus tendon can result in substantial pain. Smaller muscle groups, such as the magnus brevis and gracilis, are often traversed by this technique without substantial complications. The obturator vessels are lateral and superior to the area of insertion of the device. The dorsal nerve of the clitoris is separated from the trajectory of the device by at least 1 to 2 cm.

38. **a. Either outside-in or inside-out approaches.** The transobturator technique can be performed by insertion of the passing needles from either vaginal or obturator approaches. Associated risks of device use include leg pain, dyspareunia, and injury to surrounding structures. Cystoscopy is a useful safety adjunct and should be performed as an integral and necessary part of the transobturator MUS surgery. Different kits use different meshes, and not all meshes are similar. The kit to be used should be evaluated critically for this parameter.
39. **a. Appear to be relatively similar regardless of whether ISD is present preoperatively.** Transobturator MUS surgery outcomes are relatively similar to those seen with the retropubic slings, regardless of urethral function. Any lower urinary tract structure, including the ureter, can be injured by the transobturator trocar, including the urethra and bladder. Vaginal exposure is clearly related to mesh type. Voiding dysfunction is similar to retropubic techniques. Less urethral hypermobility probably militates against success rates with transobturator slings, such as those reported in women with higher degrees of urethral hypermobility.
40. **d. a and c.** Thigh and groin pain appear to be more commonly associated with the transobturator approach. In addition, it appears that groin pain persists longer after the transobturator midurethral slings. Most groin pain resolves after the second postoperative day.
41. **d. Obesity and diabetes are associated with fasciitis after pelvic surgery.** A review of necrotizing fasciitis in gynecologic surgery found that obesity (88%), hypertension (65%), and diabetes (47%) were all factors associated with the development of fasciitis after surgery. In their randomized controlled trial from 2002, Ward and Hilton[a] found a 2% rate of vaginal wound infection after retropubic MUS surgery. In 2010, Richter et al.[a] found a 0.7% rate of vaginal wound infection in both the retropubic and transobturator MUS arms.
42. **c. A midline vaginal incision is acceptable for patients undergoing removal of mesh that has perforated into the urethra.** For slings that perforate into the urethra an inverted-U

incision is best because this allows for exposure of the proximal urethra, bladder neck, and endopelvic fascia as well as providing a vaginal epithelial flap that avoids overlapping suture lines.

43. **e. The data support the use of autologous PVS over repeat MUS in this population.** Overall, the data guiding the choice of procedure for recurrent or persistent SUI after MUS is inadequate, with a Cochrane systematic review identifying 12 trials, but none meeting eligibility criteria (Bakali et al, 2013[a]). Both urodynamics and cystoscopy may assist in elucidating the source of failure in this patient with failure of previous anti-incontinence surgery (AUA SUI Guidelines). Sling tightening is not supported by evidence-based literature, and periurethral bulking therapy produces outcomes inferior to repeat MUS for recurrent SUI.
44. **e. Single-incision sling manufacturers are required to perform 522 postmarket surveillance studies.** In January 2012, the Food and Drug Administration (FDA) mandated that all manufacturers of synthetic prosthetic mesh and biologic graft materials marketed for pelvic organ prolapse repair and single-incision sling products perform 522 postmarket surveillance studies. Midurethral sling products (except single-incision slings) were excluded from this mandate because, in September 2011, an FDA advisory panel[a] deemed existing midurethral sling products “safe and effective.”
45. **c. Surgical mesh for transvaginal prolapse.** On January 4, 2016, the FDA issued one order to reclassify mesh used for transvaginal prolapse repair from class II device, which generally includes moderate-risk devices, to class III, which includes high-risk devices.[a] A second order requires manufacturers to submit a PMA application to support the safety and effectiveness of surgical mesh for transvaginal POP repair. The PMA pathway will require manufacturers to address safety concerns, including severe pelvic pain and organ perforation, to demonstrate safety and effectiveness. These orders do not apply to surgical mesh used for other indications, such as full-length MUS and mesh for sacral colpopexy, and remain class II devices. Single-incision mini-slings are currently undergoing 522 postmarket surveillance order studies.

Chapter Review

1. Urethral slings are the procedure of choice for the surgical correction of female SUI.
2. PVSs should be placed at the bladder neck.
3. Slings are particularly helpful in treating ISD.
4. The majority of patients who require clean intermittent catheterization after PVS placement had a neurogenic bladder preoperatively.
5. Persistent urgency or urgency urinary incontinence are more common than frank urinary retention as presenting symptoms for bladder outlet obstruction after a sling.
6. Maximum urethral closure pressure occurs at the level of the midurethra.
7. Midurethral slings are less successful in patients with a fixed urethra and/or a low abdominal or Valsalva leak-point pressure.
8. Urethral mobility before midurethral sling procedures has been shown to be predictive of success (i.e., the more the proximal urethra moves during a Valsalva maneuver, the better the cure rate for incontinence).
9. For patients with persistently elevated post-void residuals and bothersome symptoms refractory to conservative management after a sling procedure, MUS release procedures consistently provide resolution of symptoms with maintenance of continence in the majority of patients.
10. Cystoscopy is an integral part of all urethral sling procedures to visualize and exclude any injury to the urethra or bladder.
11. Periurethral bulking agents have limited success in treating SUI.
12. The use of autologous tissue for a sling has the lowest rate of erosion and infection.
13. The most common reason for patient dissatisfaction following sling surgery is the persistence or development of urgency symptoms and/or urgency urinary incontinence.
14. Synthetic material is no longer used for slings placed at the bladder neck.

15. When synthetic mesh erodes into the urethra or bladder, the mesh must be removed.
16. Obese patients with incontinence benefit from weight loss.
17. When sling release is performed, a midline incision of the sling is all that is required and the entire sling does not need to be excised.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition,* on the Expert Consult website.

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Complications Related to the Use of Mesh and Their Repair

Anne P. Cameron

Keywords

complications; pelvic pain; surgical mesh; urinary incontinence; urinary retention

Questions

1. What is a mesh erosion?
    a. When mesh gradually becomes apparent in the vagina
    b. It is an incorrect term that should not be used
    c. When mesh gradually becomes exposed in the urethra or bladder
    d. Any palpable or visible mesh in any organ
    e. A result of trocar injury to the bladder or urethra
2. Why are large pores essential for mesh to prevent complications?
    a. These make the mesh a lighter weight.
    b. They make the mesh stretch.
    c. They make the mesh cheaper to produce.
    d. They allow macrophages and other cells to penetrate the mesh.
    e. They make the mesh less visible.
3. Mesh is known to increase the long-term risk of
    a. pelvic cancers
    b. systemic (nonpelvic) cancers

c. autoimmune diseases
d. chronic fatigue syndrome
e. none of the above

4. Which of the following is a contraindication for mesh sling placement?
    a. Prior failed mesh sling
    b. Vaginal atrophy
    c. Urethral injury during procedure
    d. History of recurrent urinary tract infections (UTIs)
    e. Fibromyalgia
5. In which of the following procedures can bladder perforations occur?
    a. Retropubic sling
    b. Transobturator sling
    c. Mini/single incision sling
    d. Anterior prolapse mesh
    e. All of the above
6. In a woman with retention immediately after a mesh sling placement, which of the following treatments is contraindicated?
    a. Self-catheterization for 2 weeks
    b. Urethral dilation after 1 week
    c. Indwelling catheter for 3 days
    d. Sling loosening/division in 1 week
7. What is a desirable property for type I mesh?
    a. Low melting point
    b. Knitted
    c. Smaller pore size
    d. Clear colored material
    e. Made of polytetrafluoroethylene
8. Which of the following statements about mesh is true?
    a. New mesh exposures/extrusions cannot occur more than two years after the original surgery
    b. Conservative treatment of mesh exposures with estrogen is never effective
    c. If your technique is meticulous, you can remove the entire mesh

d. Cystoscopy always needs to be performed after any mesh sling placement
e. Dyspareunia without mesh exposure should only be treated conservatively

9. What symptom is the most likely to resolve after mesh sling revision?
    a. Urinary retention
    b. Mesh exposure
    c. Stress incontinence
    d. Urgency incontinence
    e. Dyspareunia
10. Abdominal sacrocolpopexy (ASC) mesh presenting with mesh-related abdominal abscess should be treated with:
    a. long-term antibiotics and percutaneous drainage
    b. complete abdominal and vaginal mesh removal
    c. abdominal mesh removal leaving vaginal mesh in place to avoid vaginal tissue loss
    d. vaginal mesh removal and vaginal drainage of the abscess
    e. abdominal exploration and washout
11. Which of the following statements about male mesh slings are true?
    a. The mesh perforation rate of the urethra is underreported.
    b. Retention after catheter removal should quickly be treated by sling incision to prevent bladder remodeling.
    c. If a man requires an artificial sphincter after failed mesh sling, the sling can simply be divided and left in place during artificial urinary sphincter (AUS) placement.
    d. Mesh urethral perforation is often asymptomatic.

Answers

1. **b. It is an incorrect term that should not be used.** *Erosion* is a generic term that implies that tissue has been worn away by friction or pressure. Therefore, the term should not be used.
2. **d. They allow macrophages and other cells to penetrate the mesh.** Pore size is critical to wound healing and mesh integration

by the body. Smaller pore size in woven grafts allows bacteria (1 μm) to enter but restricts entry of host macrophages and lymphocytes (50 μm), impeding immune response. When pores are large such as with polypropylene type I mesh (~ 1500 μm), fibrocollagenous tissue ingrowth can occur, which allows the mesh to integrate into the local tissue providing strength and support.

3. **e. None of the above.** With long-term use of mesh in the vagina, there has been the opportunity to follow women over many years for complications. Mesh has been hypothesized to increase cancer risk due to chronic foreign body response. This theory has been disproved with population-based studies that showed no increase in pelvic or any other cancer diagnoses in women followed for 6 years after implant. Similarly, mesh implantation has not been associated with the development of autoimmune diseases over long-term follow-up.
4. **c. Urethral injury during procedure.** Mesh perforation or later extrusion at the bladder neck or into urethra is a rare complication. It occurs due to an injury to the urethra or bladder neck during either the vaginal dissection or trocar passage. If an intraoperative urethral injury is noted during dissection for placement of the sling, the surgeon must abandon mesh placement. One should repair the urethral injury with absorbable suture and allow healing before undertaking any mesh procedure, given the extraordinary high risk of urethral extrusion of the mesh.
5. **e. All of the above.** Bladder injury is far more common with retropubic than transobturator trocars (4.8% vs. 0.6%) and is typically diagnosed on cystoscopy with a 70-degree lens during the sling placement. The trocar is seen as at the dome/lateral wall of the bladder. Mesh perforation into the bladder (0.7% to 3.5%) in the case of anterior compartment pelvic organ prolapse repair and into the rectum (0.15%) from posterior repair can present with pain, fistula formation, or, in the case of bladder perforation, recurrent hematuria, UTIs, or voiding-related pain.
6. **b. Urethral dilation after 1 week.** Urethral dilation is ineffective in the setting of bladder outlet obstruction (BOO) from an obstructing

sling and puts the patient at risk for urethral extrusion of the mesh with the dilation traumatizing the tissue over the inflexible mesh.

7. **b. Knitted.** Type I mesh is macroporous with pore size greater than 75 μm. Examples of type I mesh are polypropylene and Marlex. Marlex is a woven monofilament polypropylene. It is one of the stiffest mesh materials, with a propensity to wrinkle and a high extrusion rate. In contrast, Polypropylene is a monofilament knitted graft that is softer and more compliant than woven configurations and has larger interstices due to the loose knit (1500 vs. 600 μm) increasing elasticity. In addition, its melting point of 160°C allows it to be autoclaved. It is currently the only recommended implantable vaginal mesh for slings and pelvic organ prolapse.
8. **d. Cystoscopy always needs to be performed after any mesh sling placement.** Bladder injury is far more common with retropubic than transobturator trocars (4.8% vs. 0.6%) and is typically diagnosed on cystoscopy with a 70-degree lens during the sling placement. The trocar is seen as at the dome/lateral wall of the bladder. Complete filling of the bladder is required to evaluated for injuries as an underfilled may obscure a small perforation.
9. **b. Mesh exposure.** Mesh re-exposure if these principles are followed is near 0%.
10. **b. Complete abdominal and vaginal mesh removal.** Abdominal mesh removal is sometimes indicated after failed vaginal approach or for patients with intractable mesh-related pain, abscess/infected upper portion of the mesh, or discitis. In the case of infection, the entire mesh should always be removed since residual infected mesh after a failed partial excision causes significant morbidity, including chronic infection, sinus tracts, abscess, and fistula formation. These will require a repeat excision typically via laparotomy, and this is usually a very difficult and morbid procedure.
11. **c.** If a man requires an artificial sphincter after failed mesh sling, the sling can simply be divided and left in place during artificial urinary sphincter (AUS) placement. Sling division without actual removal is sometimes needed when placing an artificial sphincter and can be performed simultaneously without difficulty.

Chapter Review

1. One should aim to remove enough mesh to correct the problem and prevent future mesh complications. Removal of all the mesh is rarely required, and removing asymptomatic mesh is never indicated.
2. Mesh slings should be circumscribed and divided lateral to the urethra.
3. The amount of mesh removed depends on the indication: Retention = simple division; exposure/extrusion = divide and remove mesh to at least a centimeter from defect; pain = maximal vaginal removal and rarely removal of retropubic or obturator arms.
4. Bladder and urethral perforations are difficult complications to treat and can be managed endoscopically or open with the open approach being more thorough but potentially more morbid.
5. Transvaginal prolapse mesh has a higher rate of mesh exposure/extrusion than slings given the larger volume of mesh. Small exposures can be treated conservatively. Large exposures and pain are treated with vaginal mesh excision; typically the entire vaginal mesh is excised leaving only the arms in place. Vaginal pain/dyspareunia is the most difficult to treat given the multifactorial etiology of pain.
6. ASC sutures are easily removed vaginally; however, mesh is very difficult to effectively remove transvaginally, often requiring multiple attempts. Open abdominal mesh removal is a potentially very morbid procedure with high risk of bowel injury, vaginal tissue loss, and high blood loss.
7. Male slings often have temporary retention that will typically resolve over time. It is uncommon to require mesh division to resolve it, and urodynamics are helpful in determining obstruction versus bladder dysfunction as the cause.

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Additional Therapies for Storage and Emptying Failure

Timothy B. Boone, Julie N. Stewart, and Laura M. Martinez

Keywords

augmentation cystoplasty; catheterization; urethral injection therapy

Questions

1. All of the following patients would be candidates for augmentation cystoplasty EXCEPT:
    a. a patient with a neurogenic bladder and poor bladder compliance who has failed trials of anticholinergic medications and intravesical injections of botulinum toxin.
    b. a patient with a spinal cord injury and detrusor leak point pressures of greater than 40 cm H_2O and subsequent vesicoureteral reflux.
    c. a patient with refractory idiopathic detrusor overactivity.
    d. a patient with significant urinary frequency and a bladder capacity of less than 100 mL.
    e. a patient with rapidly progressing multiple sclerosis and bothersome neurogenic detrusor overactivity causing urinary leakage.
2. What is the minimum threshold for particle size of urethral injectable agents that determines migration risk?
    a. 20 μm

b. 40 μm
c. 60 μm
d. 80 μm
e. 100 μm

3. Which injectable agent was reported to migrate to pelvic lymph nodes seen in radiograph?
    a. Porcine collagen
    b. Silicone macroparticles
    c. Carbon-coated zirconium beads
    d. Bovine collagen
    e. Autologous fat
4. The Credé maneuver for emptying the bladder is relatively contraindicated in patients:
    a. with decreased outlet resistance.
    b. who are obese.
    c. with vesicoureteral reflux.
    d. with high-pressure detrusor overactivity.
    e. younger than the age of 2 years.
5. All of the following statements are TRUE regarding bladder outlet closure EXCEPT:
    a. Complete closure of the bladder neck is rarely necessary.
    b. The main indication is urethral destruction after prolonged catheter drainage.
    c. An obstructing sling or artificial urinary sphincter (AUS) is rarely feasible, if less than 1 cm of urethra exists.
    d. Reflex sphincteric activity may result in disruption of the bladder neck closure.
    e. The transvaginal approach has decreased the postoperative fistula rate.
6. Of the following statements, which one is FALSE regarding “trigger voiding” in spinal cord–injured patients?
    a. Trigger voiding can be induced by digital rectal stimulation.
    b. Reflex contractions can be generated by using somatic motor axons to innervate parasympathetic bladder ganglia cells.
    c. Rhythmic suprapubic manual pressure is usually the most effective method for trigger voiding.

d. Trigger voiding induces a reflex decrease in outlet resistance in patients with detrusor-sphincter dyssynergia.
e. Trigger voiding can be induced by squeezing the clitoris.

7. Common complications associated with a continent catheterizable channel include all of the following EXCEPT:
   a. stomal stenosis.
   b. perforation of the catheterizable channel.
   c. incontinence from the stoma site.
   d. difficulty passing a catheter through the stoma due to stricture.
   e. stomal prolapse.
8. Which of the following statements is FALSE?
   a. Chronic indwelling urethral catheterization protects against poor bladder compliance and upper tract complications.
   b. Chronic indwelling urethral catheterization compared with clean intermittent catheterization (CIC) is associated with a higher incidence of urolithiasis.
   c. Asymptomatic bacteriuria is common in catheterized patients and does not usually require treatment.
   d. Periodic upper and lower tract evaluation is important in all patients managed with chronic indwelling catheters.
   e. There is still a role for anticholinergic medications in patients managing their bladders with a chronic indwelling catheter.
9. Which of the following is FALSE regarding the use of condom catheters?
   a. Similar incidence of bacteriuria and symptomatic UTIs to indwelling urethral catheters
   b. Preferable in patients who cannot perform CIC due to lack of caregiver
   c. Use of a malleable penile prosthesis can facilitate applying and maintaining a condom catheter
   d. Condom catheters should be changed daily to decrease risk for UTI
   e. Skin maceration is more likely to occur in patients with neurogenic LUT dysfunction
10. All of the following statements are TRUE regarding catheterization EXCEPT:

a. There is no known association with intermittent catheterization and development of squamous cell carcinoma of the bladder.
b. Gross hematuria in a patient with a chronic indwelling catheter is likely related to infection or inflammation and does not require a thorough hematuria workup.
c. Urinary incontinence may worsen in patients with intrinsic sphincter deficiency who convert from an indwelling urethral to suprapubic catheter.
d. There is a lower incidence of epididymitis in men who have chronic suprapubic catheters compared with urethral catheters.
e. There is a low risk of developing squamous cell carcinoma of the bladder with chronic indwelling catheter use.

11. Incontinence-associated dermatitis (IAD) is associated with the following factors in the incontinent patient EXCEPT:
    a. it may be caused by infrequent pad changes.
    b. it is manifested by inflammation of the skin with redness and edema.
    c. it may lead to malignant lesions of the skin.
    d. it predominately occurs in skin folds.
    e. it promotes candidiasis and bacterial infections.

Answers

1. **e. A patient with rapidly progressing multiple sclerosis and bothersome neurogenic detrusor overactivity causing urinary leakage.** A patient with a progressive neurologic disease, such as multiple sclerosis, may not have the ability to perform CIC in the future and an alternative treatment plan should be considered. Noncompliance with CIC puts the patient at risk for life-threatening spontaneous bladder perforation.
2. **d. 80 μm.**
3. **c. Carbon-coated zirconium beads.** Pannek and coworkers did report particle migration (2001). This was subsequently attributed to the high pressure necessary to inject the viscous material with

large particles, resulting in material displacement into vascular or lymphatic spaces. Durasphere EXP with smaller particles is less likely to lead to this.

4. **c. With vesicoureteral reflux.** The Credé maneuver (manual compression of the bladder) is most effective in patients with decreased bladder tone who can generate an intravesical pressure greater than 50 cm H_2O and have decreased bladder outlet resistance. The Credé maneuver requires good hand control, is easier in a thin individual than an obese one, and is more easily performed in a child than in an adult. Voiding by Credé is unphysiologic, because active opening of the bladder neck does not occur, and increases in outlet resistance by a reflex mechanism may actually occur. If complete emptying does not occur, treatment to decrease outlet resistance can be contemplated, or an alternative method to empty the bladder should be used. Vesicoureteral reflux is a relative contraindication to external compression and straining maneuvers, especially in patients capable of generating a high intravesical pressure.
5. **e. The transvaginal approach has decreased the postoperative fistula rate.** Complete closure of the bladder neck is rarely necessary, because a compressive bladder neck sling is more easily performed, is less morbid, and allows transurethral access if necessary. The main indication for bladder outlet closure is urethral destruction secondary to prolonged catheter drainage in neurogenic bladder patients. A case series using “tight” autologous pubovaginal sling and lower urinary tract reconstruction for urethras destroyed by long-term Foley catheter use reported excellent results with minimal incontinence. The authors concluded that at least 1 cm of normal urethra was required for proper functioning of the sling. The risk of complications, specifically a vesicovaginal fistula, is relatively common and can be difficult to repair. It is important to remember that a bladder neck closure is much more difficult than a simple closure of the bladder wall. The bladder neck is usually hyperactive in patients with neurologic disease, and every voiding reflex includes active opening and

closing of the bladder neck, which forcibly attempts to destroy the bladder neck closure. To reduce this risk, postoperative suppression of the voiding reflex using prolonged continuous catheter drainage (3 weeks) and liberal use of anticholinergics is imperative. In addition, to reduce the risk of fistula, the repair must be watertight from the beginning, and this requires a precise mucosal closure using a running suture and multiple additional layers of muscle to reinforce the strength of the repair.

6. **d. Trigger voiding induces a reflex decrease in outlet resistance in patients with sphincter dyssynergia.** In some types of spinal cord injury or bladder dysfunction characterized by detrusor hyperreflexia, manual pressure may sometimes be used to initiate a reflexive bladder contraction—sometimes called "trigger voiding." The most effective method of initiating a reflex contraction is thought to be rhythmic suprapubic manual pressure, typically seven or eight compressions every 3 seconds. This rhythmic pressure is thought to produce a summation effect on the tension receptors in the bladder wall, resulting in an afferent neural discharge that activates the bladder reflex arc. Trigger voiding can also sometimes be induced by pulling the skin or hair of the pubis, scrotum, or thigh; squeezing the clitoris; or digital rectal stimulation. Surgical procedures to reduce outlet resistance should be considered, if significant obstruction or sphincter dyssynergia are present. In an animal model using neural rerouting, a detrusor contraction without striated sphincter dyssynergia could be initiated by scratching the skin or by percutaneous electrical stimulation in the L7 dermatome. The pathway was found to be mediated by cholinergic transmission at both ganglionic and peripheral levels. The importance of this experimental model is that somatic motor axons were able to innervate parasympathetic bladder ganglion cells and therefore transfer somatic reflex activity to the lower urinary tract.
7. **b. Perforation of the catheterizable channel.** These catheterizable channels are not free of complications, and long-term issues with catheterization, incontinence, and stomal stenosis can occur. A large retrospective study by Leslie et al. (2011) analyzed the long-term outcomes of 169 pediatric patients who had either undergone a

Mitrofanoff appendicovesicostomy or a transverse ileal, or Monti, tube. **The authors report a 39% revision rate (8% stricture, 4% prolapse, 10% incontinence, and 17% stomal stenosis at skin level).** Perforation of the catheterizable channel has not been commonly reported as a complication.

8. **a. Chronic indwelling urethral catheterization protects against poor bladder compliance and upper tract complications.** The exact etiology of upper tract deterioration in patients with long-term indwelling catheters is unclear because the bladder should be well drained by a catheter; however, it is likely related to chronic "occult" or subclinical detrusor overactivity in the face of sphincteric dyssynergy providing a functional obstruction. Regardless of the etiology, it is clinically heralded by the development of poor detrusor compliance demonstrated on urodynamic studies.
9. **a. Similar incidence of bacteriuria and symptomatic UTIs to indwelling urethral catheters.**
10. **b. Gross hematuria in a patient with a chronic indwelling catheter is likely related to infection or inflammation and does not require a thorough hematuria workup.** The long-term risk of carcinoma in the spinal cord injury patient with a chronic catheter has been estimated to be 8% to 10%. This association has not been identified in patients performing intermittent catheterization. The development of gross hematuria in patients with a chronic indwelling catheter should prompt further evaluation, including upper tract imaging, urine cytology, cystoscopy, and consideration of bladder biopsy.
11. **c. It may lead to malignant lesions of the skin.** Prolonged exposure of the skin to a wet environment may lead to supersaturation and disruption of the skin's protective barriers, thus promoting skin maceration, dermatitis, and possibly infection. IAD can be defined as inflammation of the surface of the skin with redness, edema, and, in some cases, bullae containing clear exudate. IAD predominately occurs in skin folds and may promote candidiasis or bacterial skin infections. IAD has not been associated with premalignant or malignant lesions of the skin.

Chapter Review

1. The goals of therapy for storage failure are low-pressure storage, improved bladder capacity, decreased incontinence due to detrusor overactivity, and preservation of renal function.
2. Augmentation cystoplasty is an option after failed conservative management such as pharmacologic therapy, intravesical botulinum toxin injection, or CIC.
3. Use of large bowel versus small bowel may lead to more significant metabolic disturbance, increased mucus production, and theoretically heightened risk of malignancy.
4. Goal of urethral injection therapy is to restore urethral mucosal coaptation and its hermetic seal effect, especially during periods of increased abdominal pressure.
5. Injectable agents are an option for patients who are not candidates for more invasive procedures. Efficacy and durability are generally inferior to surgery.
6. Functional urethral closure with pubovaginal sling provides pop-off valve for leakage at higher pressures and allows access for transurethral instrumentation.
7. Bladder neck closure with simultaneous LUT reconstruction may be successful in refractory patients with destroyed bladder outlet.
8. Indications for catheterizable channel include difficulty catheterizing per urethra, urethral scarring, and urethral incompetence requiring outlet closure.
9. Complications of continent catheterizable channels include long-term issues with catheterization, incontinence, and stomal stenosis.
10. Asymptomatic bacteriuria is common in catheterized patients and does not require treatment unless the patient becomes symptomatic.
11. Chronic catheterization (CIC or indwelling) may lead to poor bladder compliance and upper tract deterioration; thus periodic surveillance is recommended.
12. Credé and Valsalva maneuvers may elevate intravesical pressure enough to cause upper tract deterioration, especially in a setting of poor detrusor compliance.

13. Reflex voiding depends on the ability to stimulate detrusor contractions and may be most suitable for patients with spinal cord injury (SCI) or neurogenic detrusor overactivity.
14. Condom catheter use is associated with lower rate of bacteriuria, symptomatic UTIs, and death than indwelling Foley; also more comfortable and less restrictive.
15. Complications of condom catheter include allergic reactions, penile edema; higher risk for pressure sores in patients with impaired sensation or cognition.
16. Consumers spend billions of dollars on absorbent products; selection is often based on trial and error, cost, convenience, and manufacturers' claims.
17. IAD is inflammation of skin with redness, edema, and/or bullae with clear exudate; it is often found in skin folds and may promote candidiasis or bacterial skin infections.

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Aging and Geriatric Urology

Tomas L. Griebling

Keywords

aging; elderly; geriatrics; older adults

Questions

1. With aging:
    a. renal function increases.
    b. bladder capacity declines.
    c. hepatic function remains relatively stable.
    d. pulmonary function declines.
    e. immunologic function remains stable.
2. Urinary tract infections (UTIs) in elderly women may best be decreased by:
    a. nitrofurantoin prophylaxis.
    b. systemic estrogen administration.
    c. cranberry juice.
    d. vaginal estrogen application.
    e. α-blocker therapy.
3. In demented elderly patients, incontinence:
    a. is inevitable.
    b. is virtually always due to detrusor hyperreflexia.
    c. is unlikely to respond to therapy.
    d. is multifactorial and often reversible.

e. treatment should focus primarily on preventing skin breakdown.

4. Urinary incontinence (UI) in older people is usually:
   a. brought to a physician's attention by the patient.
   b. detected by the patient's primary physician.
   c. obvious to the urologist.
   d. detected by the physician but ignored.
   e. unknown to the patient's physician.
5. In older patients, involuntary bladder contractions:
   a. are rarely seen in asymptomatic patients.
   b. are primarily due to central nervous system (CNS) pathology.
   c. are almost always the cause of the patient's incontinence.
   d. are inevitable in demented patients.
   e. may not be the cause of the incontinence.
6. After the history and physical examination, evaluation of the incontinent older patient should include:
   a. cystoscopy.
   b. videourodynamics.
   c. postvoid residual assessment.
   d. urinary cytology.
   e. assessment of prostate size in a male.
7. Which of the following occurs as part of normal aging?
   a. Urinary incontinence
   b. A small increase in serum creatinine concentration
   c. Uninhibited detrusor contractions
   d. Increase in bladder capacity
   e. Urinary flow rate is unchanged
8. The cornerstone of treatment for persistent urgency incontinence is:
   a. behavioral therapy.
   b. flavoxate.
   c. oxybutynin.
   d. tolterodine.
   e. solifenacin.
9. Acute urinary retention in an older man:
   a. indicates the need for surgical decompression.
   b. is treated effectively with α-adrenergic blockers.

c. can be seen with detrusor hyperactivity with impaired contractility (DHIC).
d. is treated effectively with bethanechol.
e. requires treatment of the underlying urinary tract abnormality.

10. Incontinence management products (e.g., garments/pads):
    a. are reimbursed by insurance companies.
    b. should include menstrual pads.
    c. generally cost less than a dollar per day.
    d. should be chosen according to the type of incontinence rather than its severity.
    e. should be tailored to the individual.
11. The voiding diary completed by an 83-year-old woman bothered by daytime incontinence discloses 800-mL output between 8:00 am and 11:00 pm, and 1500 mL from 11:00 pm to 8:00 am. The next step should be:
    a. to have her repeat it with a record of fluid intake.
    b. to take furosemide at 7:00 pm each evening to reduce nocturnal excretion.
    c. to use pressure-gradient stockings to minimize peripheral edema.
    d. to advise her to curtail fluid intake after dinner.
    e. none of the above.
12. Anticholinergic bladder relaxants may, ironically, actually exacerbate incontinence through all of the following mechanisms EXCEPT:
    a. causing/exacerbating confusion.
    b. causing/exacerbating impaired mobility.
    c. causing/exacerbating a dry mouth.
    d. causing/exacerbating subacute urinary retention.
    e. precipitating acute urinary retention.
13. A 78-year-old woman with dementia has responded modestly to donepezil (Aricept, a cholinesterase inhibitor) for the past year. The recent onset of urgency incontinence led her primary physician to prescribe tolterodine last month while awaiting your assessment. Her incontinence has responded well. The next appropriate step is to:

a. discontinue tolterodine due to its interaction with donepezil.
b. discontinue donepezil because her cognitive function is stable.
c. discontinue both drugs because she is stable and the urgency incontinence may reflect an adverse effect of the donepezil.
d. continue both drugs and monitor her for deterioration in cognitive function.
e. taper the tolterodine.

14. A 68-year-old obese woman with significant daily stress incontinence comes for a follow-up. Her bladder diary shows maximal voided volume of 125 mL during the daytime. Each of these measures is appropriate EXCEPT:
    a. adjustment of fluid excretion and voiding intervals.
    b. advising weight reduction.
    c. teaching her postural maneuvers.
    d. consideration of surgical correction.
    e. pelvic floor muscle exercises.
15. A 72-year-old man has urinary urgency and postvoid residual (PVR) of 40 mL. He also has hypertension and aortic stenosis that has caused minimal symptoms. His friend suggested that he ask for terazosin because it helped him with similar symptoms. The most appropriate response is:
    a. to prescribe terazosin and see him again in 4 weeks.
    b. to prescribe alfuzosin instead, because it has a better side-effect profile.
    c. to obtain medical consultation before prescribing the drug.
    d. to perform urodynamic testing before deciding.
    e. to prescribe an anticholinergic agent.

Answers

1. **d. Pulmonary function declines.** Pulmonary surface area for oxygen diffusion decreases, which leads to alterations in the ventilation perfusion ratio. **With aging, renal function and renal mass decline, bladder capacity remains relatively stable but**

elasticity and contractility decline, and hepatic and immunologic function decline.

2. **d. Vaginal estrogen application.** It has been proposed that vaginal estrogens promote the growth of lactobacillus and thereby lower vaginal pH, which helps reduce pathogen colonization. Systemic estrogens are generally not prescribed in the elderly and may in fact cause incontinence in this population. Nitrofurantoin should not be given during the long term in the elderly because it may reduce renal function and lead to pulmonary fibrosis. Cranberry juice has its advocates but has not been shown to be effective in randomized trials, and α-blocker therapy in the female would not be expected to have much of an effect on residual urine as the female has few alpha receptors at the bladder neck.
3. **d. Is multifactorial and often reversible.** Incontinence is never normal, even with dementia. **Detrusor overactivity (DO) is the most common type of lower urinary tract dysfunction among demented incontinent nursing home residents, but it is also the most common dysfunction among their dry peers.** Moreover, incontinence in 40% of these individuals is not associated with DO but with obstruction (in men), stress incontinence (in women), or a combination of an outlet and a detrusor problem, and the cause does not correlate with either the presence or severity of dementia. Thus it is no longer tenable to attribute incontinence a priori to DO. **Because incontinence in the elderly is usually multifactorial, involving urinary tract as well as nonurinary tract contributions, it is often treatable.** Even among nursing home patients, studies have documented more than a 50% reduction in incontinent episodes overall and full daytime continence in nearly 40% of residents. Particularly among demented individuals, nonurinary factors are prevalent and commonly include medication use, depression, fecal impaction, UTI, atrophic vaginitis, and disorders of fluid excretion. It is important to prevent skin breakdown, but this should not be the primary approach to the incontinent nursing home resident.
4. **e. Unknown to the patient's physician.** Despite the fact that incontinence is so common and amenable to therapy, most patients

do not mention it to a physician. Reasons include embarrassment, misperception that it is a normal part of aging, belief that it is untreatable, fear of complications associated with its evaluation and treatment, or misconception that only major surgery can cure it. Moreover, when patients do mention it, most physicians either dismiss it as a normal part of aging or merely check a urinalysis. With newer undergarments and pads that better absorb and deodorize, the doctor may be unaware of the problem unless he/she asks about it.

5. **e. May not be the cause of the incontinence. It is important to realize that involuntary bladder contractions are found commonly in even continent, neurologically intact elderly; the prevalence ranges in various studies between 50% and 55%.** This fact underscores the concept that such contractions are a risk factor for UI but not necessarily sufficient. Moreover, even when such contractions are the major contributor to UI, they may be due to a urethral abnormality. More than half of obstructed individuals and approximately 25% of those with stress incontinence have associated DO that usually remits with correction of the urethral abnormality alone. The proportion of elderly individuals in whom DO remits is likely lower, but clearly it is insufficient merely to identify involuntary contractions on cystometry and attribute the incontinence to them. To be considered the cause of the UI, such contractions must reproduce the patient's type of leakage, and urethral abnormalities must be excluded. This is particularly important because a bladder relaxant medication prescribed for DO that is actually due to obstruction may precipitate acute retention.
6. **c. Postvoid residual assessment.** Determining the PVR is essential in all incontinent older individuals, not only because retention can mimic other causes of UI, but also because knowledge of the PVR will affect therapy. For instance, an older woman with DO and PVR of 250 mL would be approached differently from a woman with DO and PVR of 5 mL. The rest of the diagnostic evaluation depends on the need for diagnostic certainty. However, if surgical correction is contemplated, or if the risk of empiric therapy exceeds the benefit, further testing is warranted. Cytology is indicated when

bladder carcinoma is suspected and would be treated if found (i.e., not in a bedfast, demented patient). Cystoscopy has many indications, but it is not routinely required for evaluation of incontinence, nor is it alone sufficient to detect or exclude prostatic obstruction. Palpated prostate size correlates poorly with the presence of obstruction.

7. **c. Uninhibited detrusor contractions.** Incontinence is never part of normal aging. Even at age 90 years, at least half of people are continent. Although renal function declines in most older adults, there is no change in serum creatinine because of a concomitant and balanced decrease in muscle mass. Involuntary detrusor contractions are common in continent and even asymptomatic elderly, but are rarely seen during routine cystometry in younger people. Bladder capacity may decrease in the elderly, but there is no evidence for an increase. **Flow rate declines, not only because obstruction becomes more likely in aging men, but also because detrusor contractility appears to decrease in both sexes.**
8. **a. Behavioral therapy.** Behavioral therapy is the cornerstone of treatment for detrusor overactivity, although the type of therapy must be tailored to the individual. Bladder retraining attempts to restore a normal voiding pattern by progressively lengthening the voiding interval. Scheduled toileting aims to reduce incontinence by frequent voiding, which reduces total bladder volume and the chance of triggering involuntary bladder contractions. Prompted voiding works by regularly and frequently reminding cognitively impaired residents of the need to void. The role of medications is to supplement behavioral therapy, but only if needed. By reducing bladder irritability, such agents allow the bladder to hold more urine before the spasm occurs. However, even when continence is restored by these drugs, detrusor overactivity is still generally demonstrable. Furthermore, if the drug increases residual urine more than total bladder capacity, it may paradoxically decrease functional capacity, allowing the persistent involuntary contraction to occur at more frequent intervals. Thus before deciding that drug therapy has failed, PVR should be remeasured. Except for flavoxate, each of the agents listed has been proved effective in

randomized controlled trials that included a substantial number of elderly patients.

9. c. **Can be seen with detrusor hyperactivity with impaired contractility (DHIC).** The differential diagnosis for urinary retention extends beyond urethral obstruction, particularly in the elderly. **Patients with underactive detrusor or DHIC also may develop urinary retention. In addition, fecal impaction, pain (e.g., following hip replacement), and medications with urinary tract side effects (e.g., anticholinergics, sedating antihistamines, decongestants, and opiates) may induce acute urinary retention, particularly in patients with underlying bladder weakness or obstruction.** Thus the bladder should be decompressed for at least a week while reversible causes are addressed; the larger the PVR, the longer should be the decompression. Decompression allows some restoration of detrusor strength, which also facilitates urodynamic testing should it be necessary. α-Adrenergic blockers are effective for men with symptoms of prostatism, but clinical trials excluded patients with significant urinary retention. Bethanechol, although originally designed to improve bladder emptying in unobstructed patients, has not proved effective for this purpose (and likely not for nonobstructed patients either). Decompression in some elderly patients can reduce but not eliminate residual urine; provided it does not cause symptoms or renal compromise, subclinical retention need not necessarily be treated in all elderly patients, even if obstruction is present.
10. **e. Should be tailored to the individual.** The cost of pads is rarely covered by insurance and can easily exceed $1/day. Menstrual pads, although often used for incontinence, are usually inappropriate. They are designed to absorb small amounts of slowly leaking viscid fluid rather than rapid gushes of urine. From among the numerous types of pads and garments, selection should be tailored to the individual's needs and comorbidity; the type of incontinence matters less than severity.
11. **e. None of the above.** The patient's altered pattern of fluid excretion may occur for a variety of reasons. The most common is accumulation of peripheral edema due to venous insufficiency,

peripheral vascular disease, low albumin states (malnutrition, hepatic disease), congestive heart failure, or medications (e.g., nonsteroidal antiinflammatory drugs [NSAIDs], dihydropyridine calcium channel blockers [e.g., nifedipine], or thiazolidinediones [e.g., rosiglitazone]). Each can be readily addressed. Before doing so, however, it is important to realize that the multiple pathologic conditions so often found in the elderly may be causal, contributory, a consequence, or unrelated to the condition for which the patient seeks help. In this individual, daytime leakage is the problem. Addressing the excess nocturnal excretion will not improve the daytime problem and, if it shifts the excess nocturnal excretion to the daytime (e.g., by use of pressure-gradient stockings), therapy may exacerbate the daytime leakage. If the nocturnal polyuria can be eliminated entirely (e.g., by substituting a drug that does not cause fluid retention), this should be done. If, however, therapy will only shift excretion to the daytime, one may elect not to treat if it is not dangerous (e.g., venous insufficiency). Evening furosemide risks inducing hypovolemia and increasing her risk of falls and fracture. Daytime predominance of incontinence suggests that she has stress incontinence, or DO associated with bladder neck incompetence that is exacerbated when she is upright. Once the cause is sorted out, the appropriate intervention can be prescribed, but in this individual, it should not include alteration of fluid intake: Her daytime output is too small to contribute to her daytime UI and, unless she is ingesting 2 L after dinner, her intake is also likely unrelated to her nocturnal polyuria. Moreover, the older kidney generally takes twice as long to respond to fluid restriction as the younger one, so restricting fluid after dinner is apt to do little. This case highlights the need to tailor the evaluation and treatment to the patient rather than to a given abnormality.

12. **b. Causing/exacerbating impaired mobility.** All of the currently available bladder relaxant medications have anticholinergic properties and thus can cause anticholinergic side effects. Dry mouth (xerostomia) results from the anticholinergic effect on the salivary and parotid glands. Even the M_3-specific agents have this

effect, because M_3 receptors are the predominant receptor in these glands as well as in the bladder. Because bladder relaxants generally do not abolish the involuntary detrusor contractions, the xerostomia-mediated increased fluid intake results in the bladder filling more frequently to the volume at which detrusor contractions may be triggered. Bladder relaxants often impair detrusor contractility and can lead to subacute retention. If the increase in PVR is more than the increase in total bladder capacity, the effective bladder capacity will decrease. In turn, this could allow involuntary contractions to occur at a lower effective volume; an increase in incontinence frequency can ensue.

13. **d. Continue both drugs and monitor her for deterioration in cognitive function.** Because cholinesterase inhibitors block the metabolism of acetylcholine, there is concern that they will provoke urgency incontinence, especially in older adults who already may have underlying age-related involuntary detrusor contractions that have not yet caused incontinence. However, despite prescription of these agents to millions of demented patients, there is little evidence that they cause incontinence. Moreover, because the benefits of these drugs for dementia are modest at best and are not seen in the majority of patients who use them, patients and families may decide that the benefit of the bladder relaxant outweighs the risk. Particularly in this patient, who has already benefited from tolterodine without notable cognitive deterioration, it is worth continuing therapy and monitoring her cognitive status.
14. **a. Adjustment of fluid excretion and voiding intervals.** Recent evidence suggests that weight loss will improve stress incontinence in obese women, and data support the use of postural maneuvers, pelvic floor muscle exercises, and surgical correction as well. Adjusting fluid excretion and voiding intervals can also be useful, especially for women with volume-dependent stress leakage. It can work particularly well for women with a threshold of at least 150 mL and best in those with a threshold greater than 250 mL. However, when the threshold is this low, the extent of fluid

restriction required is usually not feasible and might even lead to dangerous dehydration.

15. **c. To obtain medical consultation before prescribing the drug.** Men with these lower urinary tract symptoms and a low PVR generally respond well to an α-adrenergic receptor blocker. However, many of these agents can reduce cardiac preload and thus impede adequate left ventricular filling and cardiac output, especially in individuals whose ventricular filling is already more difficult in the setting of left ventricular hypertrophy. The risk is exacerbated by the normal age-related decline that occurs in baroreflex sensitivity and further compounded in patients who take a β blocker and/or have aortic stenosis. Thus although the overall risks of orthostasis, falls, and fracture appear to be lower with the newer α-adrenergic agents, it would be prudent to obtain medical consultation before prescribing an α-blocker in this clinical setting. Anticholinergic therapy can be used in men with urgency and a low PVR, but because of the risk of inducing a tachycardia in a man with aortic stenosis and thereby also reducing left ventricular filling —combined with the potential risk of inducing urinary retention— prescription of an anticholinergic should not be the next step.

Chapter Review

1. Renal blood flow, renal mass, and functional reserve decrease with aging, which results in a 10-mL decrease in glomerular filtration rate (GFR) per decade.
2. Serum creatinine levels do not accurately reflect renal function in the elderly due to decreased muscle mass. The Cockcroft-Gault formula is more accurate than the Modification of Diet in Renal Disease equation for estimating GFR in the elderly population.
3. Antidiuretic hormone secretion decreases in older adults, resulting in increased nocturia.
4. With age, there is a decrease in cardiac output and stroke volume.
5. The majority of deaths in the perioperative period in geriatric patients are due to cardiovascular events; however, pulmonary problems are

the major cause of prolonged hospitalization.

6. Hepatic function and immunologic function diminish with age.
7. With aging, there is a loss of muscle mass and an increase in body fat mass.
8. Bladder capacity does not change with age; however, bladder sensation, contractility, and ability to postpone voiding decline in both sexes with age.
9. Increased involuntary detrusor contractions and decreased bladder elasticity and compliance occur with aging. Indeed, detrusor overactivity is the most common type of lower urinary tract dysfunction in incontinent elderly of both sexes.
10. There is a decrease in striated muscle in the rhabdosphincter with age.
11. Urinary incontinence, UTIs, pelvic prolapse, and bladder outlet obstruction all increase with aging.
12. Stress incontinence in elderly women is usually associated with hypermobility and some degree of intrinsic sphincter deficiency.
13. A functional assessment is correlated with health care outcomes and includes: (1) activities of daily living, (2) mobility with a slow gate speed a strong predictor of mortality, and (3) cognition.
14. Cognitive changes are frequently seen following anesthesia in elderly patients.
15. There is no difference in mortality and morbidity between general and regional anesthesia in the elderly.
16. Major geriatric syndromes include frailty, falls, pressure ulcers, multiple medications, delirium, and urinary incontinence.
17. The frailty phenotype may include an unintentional weight loss in excess of 10 lb per year, reduced grip, slowing of the gait, decreased activity, and easy exhaustion with activity.
18. Elderly men on androgen deprivation therapy are at increased risks for fractures.
19. A number of medications should be used with caution or not used at all in the elderly. For example, Demerol and prolonged use of nitrofurantoin should not be given and caution should be exercised when prescribing antimuscarinics. Anticholinergic agents are one of

the most common causes of delirium in the elderly, especially in those with preexisting cognitive or functional impairment.

20. Asymptomatic bacteriuria in the elderly does not require treatment; it does not cause incontinence.
21. Peripheral edema may be mobilized when supine and cause nocturia.
22. Conservative measures used to treat excessive fluid output at night include compression stockings, changing the time diuretics are taken or administering a rapid acting diuretic in the late afternoon, and altering the diet.
23. Poststroke fecal and urinary incontinence are not uncommon.
24. Incontinence may be due to impaired mobility and/or cognition.
25. Timed voidings may be helpful in controlling incontinence.
26. An elevated postvoid residual is common in older adults.
27. Decreased fluid consumption may worsen urge incontinence due to the concentrated urine acting as an irritant on the detrusor.
28. Pelvic floor exercises, when done correctly, may be helpful in treating incontinence.
29. If a chronic indwelling catheter is required, a suprapubic tube is preferable to a urethral catheter.
30. α-Blockers are associated with the "floppy iris syndrome," and the ophthalmologist should be informed of their use prior to any ophthalmologic surgery.
31. An underactive bladder characterized by poor bladder emptying is not necessarily due to outlet obstruction and occurs in both men and women. Both structural and functional changes occur in the bladder. Currently there is no effective medication for this condition.
32. Sleep abnormalities may be responsible for nocturia and may worsen the severity of neurologic conditions which affect the bladder.
33. Nocturia once per night in the elderly is considered normal.
34. Nocturia is usually multifactorial in the elderly and thus is often not adequately addressed with a single treatment modality.
35. Hyponatremia is a significant risk factor when desmopressin is given to the elderly.
36. Fecal and urinary problems often coexist, and one may cause the other.

37. UTIs may present atypically in the elderly with symptoms of confusion, agitation, lethargy, and anorexia.
38. Vaginal estrogens are useful for reducing UTIs in elderly women.
39. Elder mistreatment screening is an important part of the urologic evaluation, just as child abuse screening is an important part of the pediatric urologic visit.

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Urinary Tract Fistulae

Dirk J.M.K. De Ridder, and Tamsin Greenwell

Keywords

fistula; genitourinary fistula;; iatrogenic fistula; obstetric fistula; vesicovaginal fistula

Questions

1. The most common cause of vesicovaginal fistulae in the nonindustrialized, developing world is:
    a. cesarean section.
    b. surgical trauma during abdominal hysterectomy.
    c. surgical trauma during vaginal hysterectomy.
    d. obstructed labor.
    e. none of the above.
2. The most common type of acquired urinary fistulae is:
    a. vesicovaginal fistula.
    b. ureterovaginal fistula.
    c. colovesical fistula.
    d. rectourethral fistula.
    e. vesicouterine fistula.
3. Vesicovaginal fistulae (VVF) may occur as a result of:
    a. locally advanced vaginal cancer.
    b. incidentally noted and repaired iatrogenic cystotomy during hysterectomy.

c. radiotherapy for cervical cancer.
d. cystocele repair with bladder neck suspension.
e. all of the above.

4. Intraoperative consultation is requested by a gynecologist for a possible urinary tract injury during a difficult abdominal hysterectomy. There is clear fluid noted in the pelvis. The gynecologist is particularly worried about postoperative VVF formation. All of the following statements are correct regarding counseling this gynecologist EXCEPT:
   a. The incidence of iatrogenic bladder injury during hysterectomy is approximately 0.5% to 1.0%.
   b. Approximately 0.1% to 0.2% of individuals undergoing hysterectomy develop a VVF.
   c. The risk of ureterovaginal fistula is greater than the risk of VVF in this setting.
   d. The absence of blue-stained fluid in the operative field following the administration of intravenous indigo carmine does not eliminate a possibility of a urinary tract injury.
   e. All of the above are true.
5. VVF due to obstructed labor are:
   a. the most common etiology of VVF in sub-Saharan Africa.
   b. usually located at the vaginal apex.
   c. never associated with simultaneous rectovaginal fistula.
   d. typically found in multiparous women.
   e. usually smaller and simpler to repair than those associated with gynecologic surgery.
6. A 47-year-old woman presents with the new onset of constant urinary leakage 5 years after completing radiation therapy for locally advanced cervical carcinoma. All of the following may be considered part of the diagnostic evaluation EXCEPT:
   a. cystoscopy and possible biopsy.
   b. voiding cystourethrography (VCUG).
   c. computed tomographic (CT) scan of the abdomen and pelvis.
   d. urodynamics.
   e. ureteroscopy.

7. A 52-year-old woman with a history of an abdominal hysterectomy 2 months previously presents for the evaluation of a constant clear vaginal discharge since the surgery. Following oral intake of Pyridium, her pads continue to have a clear watery discharge. The most likely diagnosis is:
    a. vesicovaginal fistula.
    b. ureterovaginal fistula.
    c. peritoneovaginal fistula.
    d. vesicouterine fistula.
    e. urethrovaginal fistula.
8. In the industrialized world, postsurgical VVF are associated with ureteral injury in approximately:
    a. 0.01% of cases.
    b. 0.1% of cases.
    c. 10% of cases.
    d. 25% of cases.
    e. 50% of cases.
9. A 68-year-old woman presents with a 1-week history of vaginal leakage 6 months after completion of radiation therapy for locally advanced cervical cancer. VCUG reveals a VVF. On physical examination, the fistula is irregular and indurated, and approximately 3 mm in size. Cystoscopy reveals bullous edema surrounding the fistula, and biopsy of the fistula tract reveals only fibrosis without evidence of malignancy. There is no suggestion of recurrent malignancy on CT scan. She should be counseled that:
    a. the optimal timing for repair of this fistula may be in 5 to 6 months.
    b. the best chance to repair this fistula is with immediate surgical intervention.
    c. a vaginal approach is not indicated.
    d. the use of an adjuvant flap will not be necessary.
    e. the success rate for the repair of this fistula is similar to that of a nonradiated VVF.
10. The abdominal approach to VVF repair:
    a. is the preferred approach in all patients with VVF.
    b. has a higher success rate than the vaginal approach.

c. is suitable for the use of an omental interpositional flap.
d. is associated with less morbidity and a shorter hospital stay than the vaginal approach.
e. is more often associated with postoperative vaginal shortening and dyspareunia than the vaginal approach.

11. The vaginal approach to an uncomplicated VVF repair:
    a. is most often bolstered with use of a gracilis flap.
    b. may be accomplished with a multilayer-layer closure.
    c. requires the use of nonabsorbable suture.
    d. is not indicated for obstetric-related fistula.
    e. is contraindicated if the fistula tract is within 2 cm of the ureter.
12. Principles of urinary fistula repair include all of the following EXCEPT:
    a. excision of the fistula tract.
    b. tension-free closure.
    c. use of well-vascularized tissue flaps.
    d. watertight closure.
    e. adequate postoperative urinary drainage.
13. Level I evidence (one or more randomized control trials) exists to support which of the following statements?
    a. Preoperative administration of topical estrogens improves tissue quality prior to the repair of VVF.
    b. Preoperative administration of topical estrogens improves the success rate of transvaginal VVF repair.
    c. Preoperative administration of broad-spectrum intravenous antibiotics improves the success rate of all types of VVF repair.
    d. Suprapubic bladder drainage is superior to urethral (Foley) catheter drainage in preventing surgical failure following VVF repair.
    e. None of the above.
14. Vaginal repair of VVF is contraindicated in:
    a. multiparous women.
    b. large fistulae.
    c. radiation-induced fistulae.

d. fistulae located at the vaginal cuff.
e. none of the above.

15. Potential complications of repair for a VVF following abdominal hysterectomy include all of the following EXCEPT:
    a. stress urinary incontinence.
    b. dyspareunia.
    c. recurrence of the fistula.
    d. urinary urgency and frequency.
    e. ureteral injury.
16. Advantages of the transabdominal approach to VVF repair as compared with the transvaginal repair include all of the following Except:
    a. ease of mobilization of the omentum as an interpositional flap.
    b. decreased rate of intraoperative ureteral injury.
    c. preservation of vaginal depth.
    d. easier access to the apical VVF in individuals with high narrow vaginal canals.
    e. ability to perform an augmentation cystoplasty through the same incision.
17. Seventeen days following a transvaginal VVF repair, a cystogram is performed. The bladder is filled to 100 mL with contrast medium and several images are taken. There is no evidence of a fistula on the filling images; however, the patient was unable to void during the study. A postvoid film was not obtained. This study:
    a. demonstrates successful repair of the VVF, and the catheter should be removed.
    b. is nondiagnostic, because it was done too soon following repair.
    c. is nondiagnostic, because there are no voiding images or postvoid images.
    d. is nondiagnostic, because the bladder was not filled to an adequate volume.
    e. should be terminated and cystoscopy performed to examine for a persistent fistula.
18. Before surgical mobilization, the blood supply to a potential Martius flap (fibrofatty labial flap) is through the:

a. inferior hemorrhoidal artery.
b. external pudendal artery.
c. uterine artery.
d. inferior epigastric artery.
e. gonadal artery.

19. An interpositional flap of the greater omentum during VVF repair:
    a. may be able to reach the deep pelvis without any mobilization in some patients.
    b. is most commonly based on the superior mesenteric artery.
    c. is contraindicated in the setting of inflammation or infection.
    d. should not be divided or incised vertically in the midline because this may compromise the blood supply.
    e. is most commonly used during a transvaginal approach.
20. A 39-year-old woman presents with constant vaginal leakage for 1 month following an abdominal hysterectomy. She describes symptoms of stress incontinence before the hysterectomy. She has no urgency and is voiding normally. Physical examination demonstrates no obvious fistula tract at the vaginal cuff. Oral phenazopyridine is given, and the bladder is filled with 100 mL of saline mixed with indigo carmine. A gauze pad is packed from the apex of the vagina proximally to the introitus distally, and the patient is told to ambulate for 90 minutes. Upon the patient's return, the pad is removed and examined. The most proximal portion of the pad is stained yellow-orange, and the most distal portion is blue. This is most consistent with:
    a. ureterovaginal fistula.
    b. vesicovaginal fistula.
    c. urethrovaginal fistula.
    d. a and b.
    e. a and c.
21. Ureterovaginal fistulae are:
    a. not associated with transvaginal hysterectomy.
    b. usually associated with normal voiding patterns.
    c. best diagnosed on VCUG.
    d. found more commonly following hysterectomy for malignancy than for benign indications.

e. usually located in the middle one-third of the ureter.

22. Two weeks following an emergent cesarean section for fetal distress during labor, a 28-year-old woman reports constant leakage per vagina. Analysis of the collected fluid reveals it to have a high creatinine level consistent with urine. Physical examination, including pelvic examination, reveals absolutely no abnormalities or surgical trauma to suggest a urinary fistula. There is no stress incontinence elicited on physical examination. Renal ultrasonography demonstrates no hydronephrosis, and the bladder is empty. The most likely diagnosis is:
    a. occult vesicovaginal fistula.
    b. occult ureterovaginal fistula.
    c. urethrovaginal fistula.
    d. vesicouterine fistula.
    e. peritoneovaginal fistula.
23. Vesicouterine fistulae occur most commonly due to:
    a. low-segment cesarean section.
    b. vaginal delivery.
    c. malignancy.
    d. conization of the cervix.
    e. myomectomy.
24. Potential options for therapy of vesicouterine fistula in a patient desiring long-term preservation of fertility include:
    a. observation.
    b. cystoscopy and fulguration of the fistula tract.
    c. hormonal therapy.
    d. surgical exploration and repair of the fistula with interpositional omental flap.
    e. all of the above.
25. Two months following resection of a large urethral diverticulum extending proximally beyond the bladder neck, a patient complains of urinary leakage. All of the following may be the source of this patient's symptoms EXCEPT:
    a. a urethrovaginal fistula.
    b. a vesicovaginal fistula.
    c. stress urinary incontinence.

d. a recurrent urethral diverticulum.
e. a vesicouterine fistula.

26. Urethrovaginal fistulae in the distal one-third of the urethra:
    a. are often asymptomatic.
    b. are associated with significant bladder overactivity.
    c. cannot be repaired using a vaginal flap technique.
    d. can result in severe stress incontinence.
    e. are usually the result of malignant infiltration.
27. The most common cause of a colovesical fistula is:
    a. colon cancer.
    b. bladder cancer.
    c. prostate cancer.
    d. Crohn disease.
    e. diverticulitis.
28. CT scan findings suggestive of a colovesical fistula include:
    a. intravesical mass, air in the bladder, and bladder wall thickening.
    b. air in the bladder, bowel wall thickening adjacent to the bladder, and clear fluid in a bowel segment adjacent to the bladder.
    c. air in the bladder, bladder wall thickening adjacent to a loop of thickened bowel wall, and the presence of colonic diverticula.
    d. air in the colon, colonic mass adjacent to the bladder, and debris within the bladder.
    e. air in the colon, bladder wall thickening, and an intravesical mass.
29. In the evaluation of a possible colovesical fistula, cystoscopy:
    a. has high diagnostic accuracy in revealing the cause of the fistula.
    b. has a high yield in identifying potential fistulae.
    c. should not be performed due to the risk of sepsis.
    d. is usually normal.
    e. most commonly reveals a large connection to the bowel.
30. A 62-year-old man presents with pneumaturia and recurrent urinary tract infections. A cystoscopy is performed, revealing a bullous lesion on the posterior bladder wall. Two hours later, a CT scan is

performed, revealing air in the bladder. In this patient, air in the bladder:

a. suggests colovesical fistula.
b. may be due to a bacterial infection.
c. may be due to instrumentation.
d. is a nonspecific finding.
e. all of the above.

31. The most common cause of a ureterocolic fistula is:
    a. locally extensive colon cancer.
    b. appendicitis with an associated abscess.
    c. diverticulitis.
    d. Crohn disease.
    e. tuberculosis.
32. The incidence of rectal injury during radical retropubic prostatectomy is:
    a. 0.1%.
    b. 1.0%.
    c. 5.0%.
    d. 10%.
    e. 20-fold higher in patients undergoing laparoscopic radical prostatectomy.
33. Rectourethral fistula (RUF) formation following brachytherapy for prostate cancer:
    a. may require complex reconstructive surgery or urinary diversion for repair.
    b. is located at the level of the prostate.
    c. is associated with fecaluria.
    d. may be associated with recurrent malignancy.
    e. may relate to all of the above.
34. A 61-year-old otherwise healthy man returns to the office with symptoms of mild stress urinary incontinence and fecaluria 3 weeks following radical retropubic prostatectomy. A VCUG is performed and reveals a 1-mm fistula at the vesicourethral junction. The prostate-specific antigen (PSA) is undetectable, and the final pathology reveals organ-confined disease. This patient should be counseled that:

a. a York-Mason transsphincteric approach to this fistula is associated with a high risk of anal incontinence.
b. a trial of indwelling catheterization may result in resolution of the fistula.
c. immediate colostomy is indicated.
d. the stress incontinence will become more severe following repair of the fistula.
e. urinary and fecal diversion will be necessary to repair this fistula.

35. Pyelovascular fistulae:
    a. are usually related to percutaneous procedures in the upper urinary tract.
    b. are most often due to renal malignancy.
    c. should be treated by removal of the nephrostomy tube.
    d. usually occur following radiation therapy.
    e. are usually fatal.
36. A 74-year-old woman with a history of colon cancer and external beam radiotherapy develops ureteral obstruction, and a stent is placed. Three months later, she presents with severe anemia and ongoing bright red gross hematuria for several hours. On examination she is pale and tachycardic, with a thready pulse and a systolic blood pressure of 60. As resuscitation is initiated with fluids and blood transfusion, the next step in management is:
    a. a CT scan of the abdomen and pelvis.
    b. cystoscopy, removal of the stent, and retrograde pyelography.
    c. immediate laparotomy and possible nephrectomy.
    d. angiography.
    e. a tagged red blood cell scan to lateralize the bleeding.

Imaging

1. See Fig. 129.1. A 36-year-old woman presents with increased vaginal discharge 3 weeks after an abdominal hysterectomy. On the axial CT images in the delayed excretory phase, the most likely diagnosis is:

a. vesicovaginal fistula.
b. ureterovaginal fistula.
c. colovesical fistula.
d. ureteral duplication.
e. vesicocutaneous fistula.

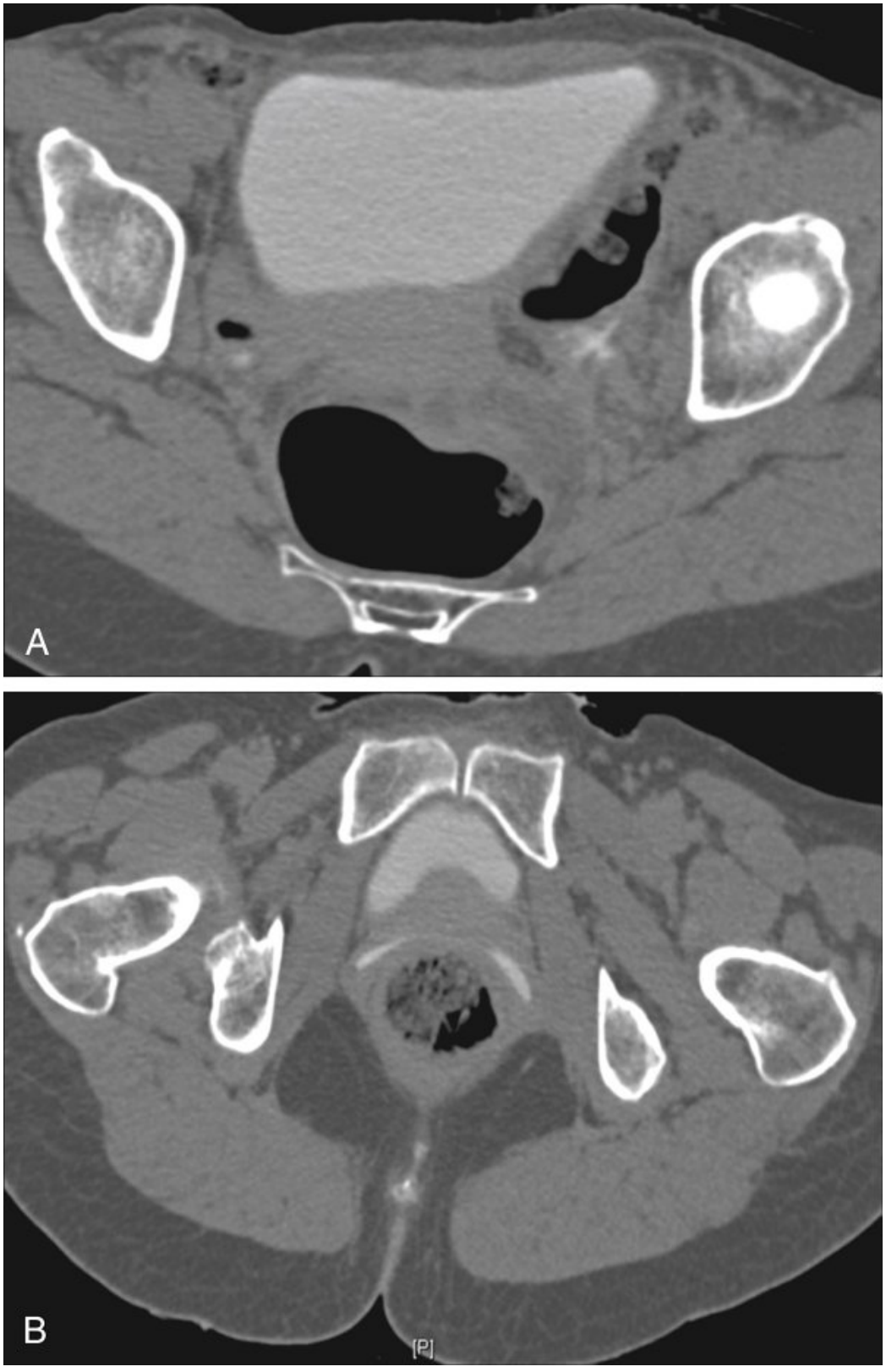


FIG. 129.1

Answers

1. **d. Obstructed labor.** In the industrialized world, the most common cause of VVF is surgical trauma during gynecologic surgery, specifically hysterectomy. In the developing world, untreated obstructed labor results in ischemic necrosis of the anterior vaginal wall and underlying lower urinary tract and is the most common fistula in these geographic areas.
2. **a. Vesicovaginal fistula.** The vast majority of urinary fistulae involve the bladder and vagina in both the industrialized and nonindustrialized world. The other types of fistulae listed are much less common.
3. **e. All of the above.** Causes of VVF in the industrialized world include surgical trauma during hysterectomy, locally advanced gynecologic malignancy, anterior vaginal wall prolapse, anti-incontinence surgery, and pelvic radiotherapy. Intraoperative recognition and repair of bladder injury during hysterectomy should reduce the probability of VVF formation, but it does not eliminate the possibility.
4. **c. The risk of ureterovaginal fistula is greater than the risk of VVF in this setting.** The most common injury to the urinary tract during hysterectomy is a bladder laceration. Although ureteral injuries are not uncommon, they occur far less frequently than bladder injuries. Furthermore, ureterovaginal fistulae are much less common than VVF. The absence of blue-colored fluid in the pelvis does not exclude injury to the urinary tract. For example, a small bladder laceration may not be evident, especially if the bladder is decompressed with a Foley catheter.
5. **a. The most common etiology of VVF in sub-Saharan Africa.** VVF in the developing world occur primarily due to obstructed labor. Typically, these occur in individuals who are young primigravidas with a narrow bony pelvis. These fistulae are usually large; located distally in the vagina, sometimes encompassing large segments of the trigone, posterior bladder wall, and bladder neck; and are often part of a larger complex of presenting signs and

symptoms termed the “obstructed labor injury complex,” which includes rectovaginal fistulae. Because of their size and extensive ischemia of the surrounding tissues, these fistulae are often difficult to repair.

6. **e. Ureteroscopy.** This individual does not have diagnosis of VVF, and therefore multiple considerations are present. Nevertheless, VVF is a strong possibility given the history of radiation therapy and pelvic malignancy. A VCUG can establish the presence of a fistula. Cystoscopy and biopsy of a fistula, if present, are mandatory to rule out recurrent malignancy. A CT scan of the abdomen and pelvis can evaluate for recurrent malignancy. Urodynamics may be helpful in evaluating for other types of incontinence, as well as assessing for bladder compliance and capacity in this individual, with a risk for impaired compliance due to radiation therapy. There is no indication for ureteroscopy in this individual.
7. **c. Peritoneovaginal fistula.** Clear fluid draining from the vagina following surgery should be properly characterized. A urinary fistula is a possible source; however, urinary incontinence (stress, urge, overflow, etc.) are strong considerations as well. A peritoneovaginal fistula is a rare complication of hysterectomy in which peritoneal fluid leaks through the vaginal cuff. The fluid may be collected and analyzed for creatinine level. A creatinine level similar to that found in serum excludes urinary fistula as the source of the fluid. In addition, if a Pyridium pad test is negative (pads are wet but are not stained orange), then this is highly suggestive of a peritoneal vaginal cuff fistula.
8. **c. 10% of cases.** Approximately 10% to 12% of individuals with VVF are found to have an associated ureteral injury.
9. **a. The optimal timing for repair of this fistula may be in 5 to 6 months.** This patient has a VVF due to radiation therapy. It is recent in onset, suggesting that the fistula is immature and has a possibility of enlarging because the radiation injury has not yet completely demarcated. The optimal timing for repair of this fistula may be in 5 to 6 months. A reevaluation at that time will be needed to assess whether the VVF is now mature and amenable to repair.

Radiation-induced fistulae can be repaired vaginally, and adjuvant flaps are used to bolster the repair. The success rates for radiation-induced VVF are less than those associated with non-radiation-induced VVF, whether they are approached vaginally or abdominally.

10. **c. Is suitable for the use of an omental interpositional flap.** The choice of approach for VVF repair is generally individualized based on the patient's anatomy, clinical circumstances, and the experience of the operating surgeon. In experienced hands, success rates are similar between the two approaches. Advantages of the vaginal approach include a shorter hospital stay and less postoperative morbidity compared with the abdominal approach; however, vaginal shortening may be an issue with some types of vaginal VVF repairs, including the Latzko operation.
11. **b. May be accomplished with a three- or four-layer closure.** The vaginal approach to VVF repair uses a three- or four-layer closure. Absorbable suture is preferred to avoid complications related to foreign bodies in the urinary tract, including stone formation and infection. Gracilis flaps are rarely necessary as peritoneal flaps or Martius labial fat flaps are much more convenient and local. The vaginal approach is not contraindicated in obstetric fistula, or if the ureter is near the fistula tract.
12. **a. Excision of the fistula tract.** Although some authors have suggested that excision of the epithelialized portion of the fistula tract is beneficial, it is not required in all cases.
13. **e. None of the above.** There is no evidence-based medicine to support any of these statements. Although both topical estrogens and intravenous antibiotics are commonly used, this is on the basis of expert opinion. There is no preferred method for postoperative bladder drainage following VVF repair, although unobstructed drainage is critical in preventing disruption of the suture line.
14. **e. None of the above.** The transvaginal approach to VVF repair can be used in most patients with uncomplicated VVF. There are few absolute contraindications to the vaginal approach. Nulliparous individuals with VVF located at the vaginal cuff in a high narrow

vagina can be challenging to repair vaginally due to anatomic considerations, but this approach is not contraindicated.

15. **a. Stress urinary incontinence.** Stress urinary incontinence may coexist with VVF; however, it is usually not related to the repair. One exception is the fistula located at the bladder neck or with involvement of the proximal urethra such as obstetric fistulae. These individuals may have new onset stress incontinence following repair due to destruction of the sphincter from the original injury.
16. **b. Decreased rate of intraoperative ureteral injury.** The transabdominal approach to VVF repair has several distinct advantages compared with the transvaginal approach. However, there are no studies to suggest that ureteral injury is less common using a transabdominal approach than a transvaginal approach.
17. **c. Is nondiagnostic, because there are no voiding images or postvoid images.** A postoperative cystogram should include voiding or postvoiding images to ensure that the VVF has been adequately repaired. Voiding may marginally increase the intravesical pressure, thereby providing opacification of some VVF that otherwise would be missed on simple filling cystograms. There is no standard filling volume for cystography. In general, 2 to 3 weeks from surgery is an adequate time period for postoperative imaging. There is no indication for cystoscopy in this patient.
18. **b. External pudendal artery.** The blood supply to the Martius flap is provided from three sources: the internal and external pudendal arteries as well as the obturator artery. In general, the small branches from the obturator artery, supplying the flap from a lateral direction, are sacrificed during mobilization. Furthermore, either the anterior (external pudendal) or posterior (internal pudendal) blood supply is divided in order to tunnel and then position the flap over the fistula.
19. **a. May be able to reach the deep pelvis without any mobilization in some patients.** The greater omentum has several favorable properties that support its use during transabdominal VVF repair. It is based on the right and left gastroepiploic arteries. Because of its rich blood supply and lymphatic properties, it can be

a useful adjunctive measure in the setting of infection or inflammation. The blood supply enters the omentum perpendicular to its origin off the greater curvature of the stomach, enabling vertical incisions and mobilization into the deep pelvis. Wide mobilization may be necessary to permit the omentum to reach the deep pelvis in some cases; however, in many individuals the flap will reach into the deep pelvis without mobilization and without tension.

20. **a. Ureterovaginal fistula.** This patient has at least a ureterovaginal fistula, based on the yellow-orange staining at the proximal portion of the gauze pad. This would be consistent with the normal voiding pattern. The distal blue staining would be consistent with stress incontinence as noted by the patient preoperatively. Hysterectomy is not associated with formation of urethrovaginal fistula. Vesicovaginal fistula is less likely because the staining would tend to be green (a combination of blue and yellow) and located in the midportion of the pad. A VCUG would be most helpful in definitively ruling out a vesicovaginal fistula.
21. **b. Usually associated with normal voiding patterns.** Ureterovaginal fistulae involve the distal one-third of the ureter. They most commonly occur in the setting of hysterectomy: laparoscopic, abdominal, and vaginal hysterectomy may all result in ureterovaginal fistulae. Most ureterovaginal fistulae occur following hysterectomy for benign indications. Patients often do not complain of voiding dysfunction because the contralateral upper urinary tract provides filling of the bladder. VCUG is used primarily to exclude a concomitant VVF.
22. **d. Vesicouterine fistula.** The most common cause of vesicouterine fistula is low-segment cesarean section. The normal physical examination suggests a lack of surgical trauma to the vagina, which most likely excludes a vaginal fistula. In the postpartum period, urine from a vesicouterine fistula will leak out of the incompetent cervical os, resulting in constant urinary leakage. A VCUG will confirm the diagnosis.
23. **a. Low-segment cesarean section.** The vast majority of vesicouterine fistulae occur following low-segment cesarean

section. Rarely, these may occur due to uterine rupture at the time of vaginal delivery.

24. **e. All of the above.** All of the listed options may preserve long-term fertility in patients with vesicouterine fistula. In those not desiring preservation of fertility, hysterectomy is indicated.
25. **e. A vesicouterine fistula.** It is very unlikely that a vesicouterine fistula can result from such a clinical circumstance. Stress incontinence, VVF, urethrovaginal fistula, and a recurrent diverticulum may all result in the described symptoms.
26. **a. Are often asymptomatic.** Distal urethrovaginal fistulae are often asymptomatic, because they originate beyond the sphincter. Vaginal voiding and pseudoincontinence may be present in some patients. A vaginal flap technique is an effective method of repair.
27. **e. Diverticulitis.** Diverticulitis is the most common cause of colovesical fistula in most series. Colon cancer is the second most common cause, followed by Crohn disease.
28. **c. Air in the bladder, bladder wall thickening adjacent to a loop of thickened bowel wall, and the presence of colonic diverticula.** The classic triad found on CT scan, which is suggestive of a colovesical fistula, includes air in the bladder, bladder wall thickening adjacent to a loop of thickened bowel, and the presence of colonic diverticula.
29. **b. Has a high yield in identifying potential fistulae.** The finding of bullous edema during cystoscopy is nonspecific; however, in the appropriate clinical setting, this can be very suggestive of a colovesical fistula. Eighty percent to 100% of cases of colovesical fistulae have an abnormality noted on cystoscopy. Cystoscopy and biopsy are useful to rule out a malignant fistula when this is a consideration.
30. **e. All of the above.** Air can be introduced into the bladder from instrumentation (i.e., cystoscopy or catheterization) or may be present due to infection with a gas-forming organism. Less commonly, air in the bladder results from a colovesical fistula.
31. **d. Crohn disease.** Most ureterocolic fistulae occur on the right side and occur in patients with Crohn disease. Left-sided fistulae in Crohn disease are much less common.

32. **b. 1.0%.** Most large series report a 1.0% to 1.5% incidence of rectal injury during radical retropubic prostatectomy. When recognized and repaired intraoperatively, very few of these injuries result in a rectourethral fistula. The incidence of rectal injury during laparoscopic radical prostatectomy, when performed by experienced surgeons, is similar to that reported in most open series.
33. **e. May relate to all of the above.** RUF commonly present with fecaluria, regardless of the etiology. RUF in the setting of prostatic malignancy should be biopsied to evaluate for the possibility of recurrent disease.
34. **b. A trial of indwelling catheterization may result in resolution of the fistula.** This is a small fistula, and as such, a trial of conservative therapy is warranted. Because this fistula is not associated with signs of local infection or sepsis, immediate colostomy is not indicated. A York-Mason operation is not associated with a high rate of anal incontinence. Furthermore, a single-stage approach may be attempted (without fecal diversion) in this uncomplicated fistula, if conservative measures fail. Finally, urinary incontinence may not worsen following surgical repair of the fistula.
35. **a. Are usually related to percutaneous procedures in the upper urinary tract.** Pyelovascular fistulae are most often related to interventional procedures in the upper urinary tract, especially percutaneous procedures. Renal neoplasms and radiation therapy are not usually causative of these fistulae. Initial treatment consists of tamponade of the bleeding vessel. If this is unsuccessful, angiographic embolization may be necessary.
36. **d. Angiography.** This individual is at high risk for a ureteroarterial fistula at the level of the stent. A CT scan and retrograde pyelography will both most likely be nondiagnostic. Removal of the stent could result in an increase in bleeding and be rapidly fatal. Angiography in the setting of active bleeding will provide both the diagnosis of a ureteroarterial fistula, if present, and a possible therapeutic intervention in the form of embolization or stent graft placement. Nephrectomy will not stop the acute hemorrhage. A red

blood cell scan will be too time consuming, and although it may lateralize the side of the bleeding, it will delay a potentially lifesaving intervention.

Imaging

1. **b. Ureterovaginal fistula.** There is extraluminal contrast around the left distal pelvic ureter with contrast opacification of the vagina on the lower image. The bladder is normal in appearance with no contrast extravasation, making options a, c, and e incorrect. Ureteral duplication does not have this appearance.

Chapter Review

1. Vesicovaginal fistulae may occur many years after completion of radiation therapy.
2. Clear vaginal discharge may not invariably represent a urinary fistula but may be a sign of a peritoneal vaginal fistula, lymphatic fistula, vaginitis, or fallopian tube fluid.
3. A fistula that does not heal following primary repair should be suspected of being associated with poor nutrition, a fungal infection, a malignancy, tuberculosis, distal obstruction, or the presence of a foreign body.
4. In the repair of fistulae, multiple layers should be used, and there should be no overlapping suture lines.
5. Long-term complications of vesicovaginal fistula repair include vaginal shortening and stenosis.
6. For an abdominal repair of a vesicovaginal fistula, it is essential to mobilize the bladder caudal to the fistula. Cholinergic agents are used liberally in the postoperative period following repair of a vesicovaginal fistula.
7. A Martius flap may be divided at either its superior or its inferior margin, because the vascular supply is provided at both ends of the graft.

8. A peritoneal flap is mobilized without opening the peritoneum, advancing it and securing it in a tension-free manner between the bladder and the vagina.
9. Following a ureteral injury, decompression of the upper tracks is essential.
10. Vesicouterine fistulae do not always present with urinary incontinence.
11. Soft tissue flaps are an important component of successful urethrovaginal fistula repair.
12. A recurrence of the malignancy should be ruled out in any fistula that develops following treatment of a primary malignancy with radiation therapy.
13. Vesicovaginal fistulae following hysterectomy are usually located on the anterior vaginal wall at the vaginal cuff.
14. Tissue interposition should be considered when repairing a fistula that failed primary closure, very large fistulae, and those occurring following radiation therapy.
15. The gracilis muscle, the rectus abdominis muscle, and a Martius pad are excellent flaps for tissue interposition.
16. An endovascular stent should be considered for ureterovascular (usually iliac) fistula repair.
17. Distal urethrovaginal fistulae are often asymptomatic because they originate beyond the sphincter.
18. Diverticulitis is the most common cause of colovesical fistula in most series. Colon cancer is the second most common cause, followed by Crohn disease.